

Comparative Size of Brains and Brain Components

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INTRODUCTION

The functional capacity of a biological system in general, and of the brain in particular, depends on its size and structural differentiation. Small and poorly differentiated brains and brain components are not as capable of such highly developed and complex functions as those that are larger and better differentiated. As a rule, the two variables (size and differentiation) do not vary independently. An increase in size is almost always accompanied by a progressive differentiation, and vice versa. However, the differentiation can vary in many ways (eg, in construction, arrangement, and connections of cells, units, layers, etc), whereas there are only two possibilities for variation in size: enlargement or reduction. Thus, size comparison may serve as a relatively simple first criterion for the functional capacity of a brain or brain part in a given species.

The questions of the most appropriate method for comparing brain size and evaluating the results obtained are still open to controversy. Our requirements for a method are that it 1) be easily applicable to large numbers of specimens and species, 2) give similar results for species that are closely related and thus have similar basic morphological and ecoethological features, and 3) reflect differences in these features in different taxonomic groups in an intuitively reasonable manner (eg, high, possibly highest position in *Homo sapiens*).

The purposes of this chapter are to examine methods of scaling brain size and select the best available; to compare, by means of this method, the size of brains and brain parts of various primate species and larger taxonomic units; and to

stress major trends in brain evolution and interpret size differences in behavioral or ecoethological terms, since differences in the size of the brain and its parts should reflect major evolutionary trends in niche diversification.

PROBLEMS IN SCALING OF BRAIN SIZE

An easy way is to scale absolute brain size, eg, the brain weight (BrW). Within extant primates, BrWs vary from 1.78 g in *Microcebus murinus* to 1.33 kg in man (Table 1). The ratio between the smallest and the largest brain is thus about 1:750, and the ratio between the smallest and any given brain could be used as a relative measure. However, such ratios are biased in that the larger ratios (brains) are found in the larger species and the smaller in the smaller. Closely related species of different body size have different ratios, and, in mammals, large whales have much higher ratios than man. To compensate for the strong influence of body size and to obtain a more independent measure of relative brain size (RBrS), the body weight (BoW) was taken into consideration by its ratio with BrW [eg, Cuvier, 1845; Brandt, 1867; Weber, 1896]. However, in this method, the RBrS values (ratios or percentages) are in general high in small species (4.17% in *Galagoides*) and low in large species (0.48% in *Gorilla*, 2.05% in man). Such differences exist even in closely related forms, and man has a lower RBrS than small prosimians. Therefore, these two methods do not meet the requirements.

That the RBrS values progress in two opposite directions when going from small to large species, ie, either in an ascending order when BoW is

ignored ($BoW^0 = 1$) or in a descending order when BoW is fully taken into account (BoW^1) suggests that the power that fulfills the requirements of similar RBrS in narrowly related species and possibly the highest RBrS in man lies between 0 and 1. Snell [1892a,b] used a power of 2/3. He introduced the general power function ($y = bx^a$) which later was extended by Klatt [1919, 1921], Huxley [1932], and Teissier [1936] to the wide range of phenomena that encompass differential or allometric growth [Röhrs, 1959; Gould, 1971]. The two methods mentioned above can easily be transformed to this power function in which $x = BoW$; a , the BoW power; $y = BrW$; and b , the measure of RBrS:

$$BrW = RBrS \times BoW^a \text{ and } RBrS = \frac{BrW}{BoW^a}$$

When BoW is ignored, a is 0; when BrW is calculated as a percentage of BoW, a is 1. The power function represents a parabola, which in its logarithmic transformation plots as a straight line in which $\log RBrS$ is the y intercept, and the power of a is the slope of the line:

$$\log BrW = \log RBrS + a \times \log BoW.$$

A power function of this type has proved to be the best method for determining lines that may serve as reference lines for evaluating RBrS values that meet the stated requirements. The key is a careful determination of the slopes, and this item will therefore be discussed in detail below. Before doing so, the measure of brain size and the standards of comparison will be considered. Finally, the position of the reference line that would give meaningful RBrS values is discussed.

Measure of Brain Size

The size of the brain and its parts is usually expressed by weight or volume. One- or two-dimensional measures are sometimes used to solve special problems, such as the extent and thickness of laminated structures (eg, neocortex) or of cross sections of fiber bundles (eg, corpus callosum). In cortices and fiber bundles, size may be better characterized by such measurements. However, for a general comparison including both cortical and noncortical structures, volumes (weights) are the most universal measure, since linear and surface measures of subcortical gray matters are, in gen-

eral, meaningless. Estimates of cell numbers are valuable; however, all cells cannot be counted except in very small structures, whereas in larger structures, cell numbers can be determined only by extrapolating findings from random samples to the volume of the structure under investigation.

Standards of Brain Size Comparison

RBrS is usually considered as brain size relative to BoW. However, many attempts have been made to replace BoW by other standards of comparison as well, either because 1) BoW simply is not known, or, more fundamentally, 2) BoW is not expected to be a useful standard. The rejection is often justified by the fact that BoW may be influenced by individual variations due to actual food ingestion, age, gravidity in females, captivity, etc. Therefore, to avoid such difficulties, brain size comparisons have been made relative to body length or to the size of a particular brain part, such as the size (expressed as weight, volume, cross-sectional area, length) of the medulla oblongata [Anthony, 1938; Passingham, 1975; Hofman, 1982] and the size (weight) of the brainstem [Portmann, 1942; Wirz, 1950]. Other authors have used the weight of the spinal cord [Krompecher and Lipak, 1966] or the size of the foramen magnum [Radinsky, 1967] as the standard of comparison. Foramen magnum size was used by Radinsky when skulls and endocranial casts were available but body size was not known—eg, in extinct species.

Like BoW, the substitutes also pose problems. Besides the reliability of such substitutes in reflecting body size, their practicability for large numbers of specimens must also be questioned.

Body length is easier to obtain than BoW, especially in big animals (eg, whales). However, 1) as a one-dimensional measure, its degree of accuracy is much lower than that of BoW, 2) several investigators may obtain fairly different results when measuring the body length of the same individual, and 3) body length depends on shape. Long and slender forms do not have larger brains than comparable short and compact forms [Stephan, 1954; Gould, 1975].

Substitutes for the BoW taken from the brain itself either are difficult to obtain and thus less useful for large numbers of specimens (eg, the volume of the medulla oblongata), are less precise (eg, the weight of the brainstem when macroscopically dissected), or only vaguely reflect body size (eg, foramen magnum size). Furthermore, these substitutes are not independent of brain size, and they may be strongly influenced by specializa-

tions, such as water adaptation [Stephan and Dieterlen, 1982; Stephan and Kuhn, 1982].

Despite intraspecific variation of BoW, which has led investigators to replace it by other standards, BoW will be used in this chapter. The main practical reasons are that most of the other potential standards are obtained with much greater difficulty and that they have weaknesses. We believe that, of all the easy-to-use parameters, BoW is the most reliable as a standard for BrW comparisons.

BoW is the sum of the weights of all the parts of the body and thus also includes BrW. However, the influence of BrW on BoW is limited since BrW never seems to exceed about 4% of BoW. It is highest in very small shrews (*Microsorex* 3.8%) and in relatively small primates (*Galagoideus* 4.2%). In man, slightly more than 2% of the BoW is BrW. Thus the measurable influence of absolute BrW on absolute BoW is negligible.

The Appropriate Best-Fit Line and Its Slope

Methods of Estimation. The use of the power function ($BrW = RBrS \times BoW^a$) is expected to be the best way to determine a reference line for evaluating RBrS. Since the data points of the selected samples of species are not really on a straight line, it is necessary to estimate a best-fit line. However, there is little agreement on the question of the most appropriate best-fit line. Three such lines are in common use: the least-squares regression line [eg, Huxley, 1932], the major axis, and the reduced major axis [Kermack and Haldane, 1950; Jolicoeur, 1965]. The least-squares regression analysis has often been rejected as inappropriate for studies of allometry, because it assumes that only one of the two variables is subject to error or fluctuation. Since allometry refers to the relationship of two morphologic variables that are assumed to be subject to independent random fluctuation, the least-squares regression analysis does not seem to be an appropriate estimation procedure [Kuhry and Marcus, 1977]. However, as was pointed out by Steudel [1982], "a fundamental aspect of allometric research is asking how a structure or variable changes with a change in body size," and therefore, BoW should be regarded as the independent variable. Fortunately, in practice, the three procedures give very similar results when the correlation is high—ie, when there is little scatter of the points around the lines. In this paper, the least-squares regression method is used for determining the best-fit line. Different slopes have been found depending on the taxonomic level of the sample.

Sample Diversity. One group of authors has used, for the calculation of the best-fit line, samples of mammals belonging to several orders and found a slope around 0.66 (0.655 [Bonin, 1937]; 2/3 [Jerrison, 1973]). The use of larger samples in recent studies has led to estimation of steeper slopes (0.74 [Eisenberg, 1981]; 0.755 [Martin, 1981, 1982]; 0.732 [Hofman, 1982]; 0.72 [Armstrong, 1982]; 0.761 [Armstrong, 1983]). Evidently, such slopes across orders depend strongly on the mixture of the combined taxa, which frequently seems to be random.

Other authors have based their estimation of the slope on lower taxonomic levels, such as genera, families, or orders. These estimated slopes show a wide range of variation with a clear tendency to be lower than 0.67, although higher values have sometimes been found. Snell [1892a,b] expected the "somatic exponent" of closely related species to be 2/3. Dubois [1897] and later Brummelkamp [1940] found it to be about 0.56 for related species. For various mammalian groups, Wirz [1950] calculated slopes ranging from 0.22 to 0.85. For Insectivora and Primates, the slopes of the regression lines vary between 0.37 and 0.75 [Stephan and Andy, 1964; Bauchot and Stephan, 1966, 1969]. By comparing the slopes of as many related groups as possible in Insectivora, Primates, and Chiroptera, these authors found an average of about 0.63. This was considered the most typical size relationship between BrW and BoW [Stephan, 1972]. More recently, Clutton-Brock and Harvey [1980] calculated slopes of the best-fit line for seven primate families. Three families with more than ten congeneric groups had slopes varying from 0.532 to 0.696, whereas families with fewer than five congeneric groups had either relatively steep (0.78) or exceptionally shallow slopes (<0.2). Contrary to Sholl [1948], who found a slope of around 0.25 for the genus *Macaca*, Bronson [1981] found a slope of 0.61.

In intraspecific comparisons of the relationship of BrW and BoW, much lower slopes are usually found than in interspecific comparisons. Dubois [1897] and Lapique [1907a,b] accepted a general value of 0.26, and Brummelkamp [1940] accepted a value of 0.22. Different values for different species were found by Klatt [1921]. According to Sholl [1948], who used various ways of calculating the best-fit line, the values for different *Macaca* species vary from 0.017 to 0.173. For dogs, Stephan [1954] found a slope of 0.25; for Insectivora, Bauchot and Stephan [1964] found values ranging from 0.18 to 0.36. To compare the brain size of

various primate species, Hemmer [1971] used a slope of 0.23. Leutenegger [1973] used the same slope to calculate brain size indices in australopithecines.

Assessment of the Appropriate Taxonomic Level. The preceding review shows that in BrW/BoW relations, the slope of the intraspecific lines is generally lower than 0.4, whereas the slope of interspecific lines is said to be as high as about 0.8, or even higher. As a consequence of their low slope, intraspecific lines appear to be one above the other when plotted along the regression line for a family [see Bauchot and Stephan, 1964]. Similarly, interspecific lines for lower taxonomic groups, such as genera, families, or subfamilies, appear as stepwise increments along the lines within or across orders.

The consensus seems to be that the slopes for lower taxonomic units (closely related species) are usually less steep than those for higher taxonomic units [Passingham, 1975; Clutton-Brock and Harvey, 1979a; Lande, 1979; Martin, 1982]. Our data, and also those of Bronson [1981], however, show that at least on the genus, subfamily, and family levels, the slopes are similar [see also Radinsky, 1982].

The decision as to which taxonomic level is the most appropriate depends mainly on the purpose of the comparison. Our purpose is to evaluate species-specific differences in the relative size of brains and brain parts. Within a population (species), BrW variation is considered a correlative adjustment to BoW to maintain the normal function of the organism. Thus, an intraspecific line expresses nothing more than the variation of BrW within a population of individuals with BoW differences. Furthermore, the lower coefficient of variation of adult BrW, which is roughly half that for BoW, indicates that intraspecific variation of BoW depends heavily on environmental factors [see Lande, 1979]. Intraspecific brain allometry is therefore irrelevant to the question of species-specific changes in BrW in relation to species-specific changes in BoW.

On the other hand, it is difficult to see the real biological implication of the residuals of allometric studies with regression lines fitted for heterogeneous groups, such as samples including representatives of various or even all mammalian taxa, because those lines do not express changes in BrW solely coupled to changes in BoW but rather express the general trend in the increase of brain tissue for the sample under consideration.

The most meaningful sample for evaluating the best-fit line that expresses species-specific changes in BrW in relation to species-specific changes in BoW consists of closely related species with similar body shape but different BoW. In this case, we may assume that brains of different BrW are functionally equivalent in terms of survival value. The increase in BrW not only corresponds to the increase in somatic tissue of the animal but also reflects a change in the ecological niche due to the increased BoW.

The dilemma caused by the uncritical use of either intraspecific slopes or interspecific slopes based on high taxonomic levels (heterogeneous groups) has been demonstrated recently by the controversy between Leutenegger and Walker et al about the encephalization in *Proconsul africanus* [Walker et al, 1983; Leutenegger, 1984; Smith and Walker, 1984].

The Constancy of the Slope. When using lower taxonomic units (genera to families), there is yet another important prerequisite for applying the power function $\text{BrW} = \text{RBrS} \times \text{BoW}^a$ for broader comparisons: the need to use a uniform slope to evaluate differences in RBrS. This was done from the outset: Snell [1892a,b] introduced a constant exponent of 2/3, and Dubois [1897] one of 0.56. In log-log plots, constant slopes are figured by parallel lines. The distances between them are equivalent to the differences between the b values (y intercepts).

The use of the power function (in its log-log form) as a method for broader comparisons and the b values as a measure of RBrS is only possible if one considers the slope constant. Then, the point at which the distances between the parallel lines (ie, the differences in RBrS) are measured becomes irrelevant. They may be expressed as the differences between their y intercepts (when $\log \text{BoW} = 0$) or at the level of the log of their actual BoW (measured parallel to the ordinate) and may be expressed in logs or antilogs, in ratios or indexes. Each of these measures is an evaluation of RBrS, and the choice is only a question of clearness, intelligibility, or general applicability.

Evaluations can be made not only of the distance between two parallel lines but also of the distances of single points from a reference line. By this means, single species or groups of related species whose common slopes may differ from that of the reference line may also be included in a comparison.

son. However, it is not possible to compare two samples with different slopes using their individual y intercepts: a group with larger brains but a steeper slope may have lower y intercepts, and thus smaller RBrS, than a group with smaller brains but a shallower slope. This may happen even when both groups have similar BoWs.

Choice of a Satisfactory Position for the Reference Line

In contrast to the difficulties in determining a common slope, the position of the reference line poses few problems. The placement depends on the aim of the comparison and may thus be determined by phylogenetic, ecoethological, or physiological considerations. The reference line can easily be displaced by parallel shifting from one reference group to another. The distances between the log-log points, and thus ratios of RBrS of different species, are independent of the position of the reference line.

Many authors use as a reference line the best-fit line for all mammals which predicts the brain size of average mammals for any given body size. Jerison [1973] called the deviation of a particular brain size from this line the encephalization quotient (EQ). The comparative brain size indexes (CBS) proposed by Clutton-Brock and Harvey [1980] are used to compare "congeneric groups"—ie, congeneric species that "share the same ecological category and social system" within a family. They express the distance of the mean BrW of a congeneric group from the best-fit line for the family.

One of the major problems in evolutionary studies is to decide which characters are primitive or advanced, which are progressive or regressive. Previous studies on brain evolution in primates and other mammals have shown that meaningful results can be obtained by determining RBrS based on a primitive group [Stephan and Andy, 1964; Stephan and Bauchot, 1965; Bauchot and Stephan, 1966, 1969; Stephan, 1967a,b; Pirlot and Stephan, 1970; Stephan and Pirlot, 1970; Jolicoeur et al, 1984]. Since it is impossible to obtain precise information on the quantitative composition of the brain in primitive fossil mammals, the most primitive forms of extant Insectivora are used as a reference group ("basal" Insectivora). Their brains are expected to be similar to those of extinct forms that are considered forerunners of many mammalian orders, including primates. Brain sizes of var-

ious species in relation to those of isoponderous primitive Insectivora have been investigated since 1956 by Stephan and co-authors. RBrS was expressed as indexes of encephalization (EIs) [eg, Stephan and Bauchot, 1965; Bauchot and Stephan, 1966, 1969].

The choice of a primitive group as the main reference group does not preclude using another taxonomic unit (eg, *Homo* [Holloway and Post, 1982]) as a reference group for intragroup as well as for intergroup comparisons. This is particularly relevant when intragroup diversity of brains and brain parts in restricted comparisons is considered (see above), or when a brain structure is not sufficiently differentiated in the "basal" group, to provide precise measurements (as, eg, the visual cortex in Insectivora).

Summary

Our requirements for a method for estimating values of the relative size of brains and brain parts in larger numbers of specimens and species are to give 1) similar values for narrowly related species, 2) possibly highest values for man, and 3) differences in values that are compatible with known data on predominance of sensory systems, behavior, and ecology.

The brain size is best expressed by brain weight (BrW) or volume; its interrelationship with body size (body weight, BoW) is best represented by the power function $\text{BrW} = \text{RBrS} \times \text{BoW}^a$. In log-log plots, RBrS—ie, the measure of relative brain size—is the y intercept. Meeting the requirements given above depends on the power "a" of the BoW, which differs with the level of the taxonomic units used for its calculation (least-squares regression method in this paper). The power a is the slope of the regression line of a sample (log-log plots). For broader comparisons including all Primates (and Insectivora), it is appropriate to use a constant slope, which may be selected from the greatest possible number of samples. When using a constant slope, the position of the reference line is of minor importance. We prefer lines through primitive forms of extant Insectivora.

COMPARISON OF TOTAL BRAINS (ENCEPHALIZATION)

Brain Weight Comparisons in Isoponderous Species

In isoponderous species, the ratios of the BrWs are the same as those of the RBrS values. There-

fore, in such species the RBrS values are easy to determine by calculating the ratios of the BrWs. We selected five groups of equal or similar BoWs from our data (around 115 g, 250 g, 850 g, 6 kg, and 56 kg) and found that in the three groups, which included Insectivora (all groups of up to 1 kg), the smallest BrWs were always those of Tenrecinae (Fig. 1). The prosimian BrWs were found to be 3.0–4.9 times larger than those of isoponderous Tenrecinae, the BrWs of Callitrichidae (present in 2 groups), 5.0 times larger, and the BrWs of Cebidae (850 g group only), 6.7–7.4 times larger. In this last group, the BrWs of monkeys were on the average 1.8 times larger than those of prosimians. In the group with BoW of about 6 kg, the smallest BrW was found in the *Indri*. The BrWs of the monkeys (*Alouatta*, *Cercopithecus*, *Presbytis*) were between 1.4 and 2.0 times (average 1.7 times), and those of apes (*Hylobates*), 2.5 times larger than that of the *Indri*. In the 56-kg group, comprising only apes and man, the BrW of *Pan* was about 1.1, and that of man, about 3.5 times larger than that of *Pongo*.

Within the three groups of up to 1 kg BoW, the prosimians had on the average similar ratios as compared to tenrecs; the ratios were 4.3 ($n = 1$), 3.8 ($n = 2$), and 3.9 ($n = 4$). If we assume that this would also be true of larger prosimians, we

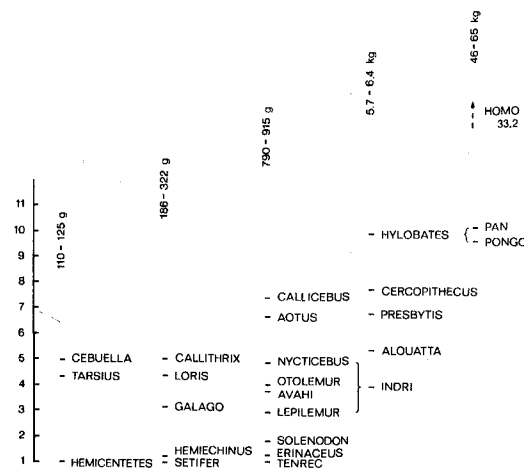


Fig. 1. Direct comparison of BrWs in species with equal or similar BoWs (isoponderous groups). The left-hand scale shows how many times the BrWs of the various species are larger than the smallest BrW within a given group. This refers to the first three groups only. The two last (heaviest) groups are adjusted to the results of the first three groups.

can expect a ratio of about 3.9 for the *Indri*. Then, in comparison with Tenrecinae, the BrWs of the other members of the 6-kg group would also be 3.9 times larger; the BrWs of monkeys would be between 5.3 and 7.6 (average 6.8), and that of *Hylobates*, 9.8 times larger than the BrWs of isoponderous tenrecs. With reference to *Hylobates*, the BrW of man (1,330 g), being 3.4 times larger than the average of pongids of similar BoW, would

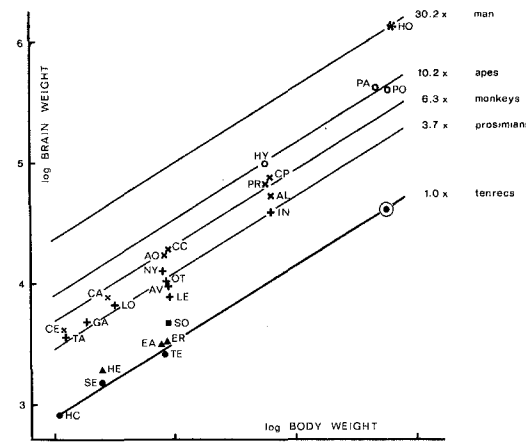


Fig. 2. BrWs (in milligrams) plotted against BoWs (in grams) in a double logarithmic scale. The species are the same as presented in Figure 1. The connecting line between three existing species of Tenrecinae and the theoretically derived large tenrec has the equation $\log \text{BrW} = 1.638 + 0.624 \times \log \text{BoW}$. Thinner parallel lines are through the averages of prosimians ($n = 8$), monkeys ($n = 5$), and pongids ($n = 3$) and through the point for man. BoWs and BrWs are given in Table 1. BoW/BrW data from species not included in Table 1 are given in parentheses after the species' names. Abbreviations: AL, *Alouatta seniculus*; AO, *Aotus trivirgatus*; AV, *Avahi laniger occidentalis*; CA, *Callithrix jacchus*; CC, *Callicebus moloch*; CE, *Cebus pygmaea*; CP, *Cercopithecus mitis*; EA, *Erinaceus algericus* (790/3,200); ER, *Erinaceus europaeus* (860/3,350); GA, *Galago senegalensis*; HC, *Hemicentetes semispinosus* (110/830); HE, *Hemiechinus auritus* (250/1,900); HO, *Homo sapiens*; HY, *Hylobates* spp. (5,800/96,000); IN, *Indri indri*; LE, *Lepilemur ruficaudatus*; LO, *Loris tardigradus*; NY, *Nycticebus coucang*; OT, *Otolomur crassicaudatus*; PA, *Pan troglodytes*; PO, *Pongo pygmaeus* (58,000/383,000); PR, *Presbytis obscura* (5,800/66,000); SE, *Setifer setosus* (248/1,510); SO, *Solenodon paradoxus* (900/4,670); TA, *Tarsius* spp.; TE, *Tenrec ecaudatus* (832/2,570).

be about 33.2 times larger than that of a hypothetical tenrec of equal BoW. Thus, a hypothetical tenrec (as a member of the 56-kg BoW group) would have a BrW of about 40.1 g (1,330:33.2).

As shown in the preceding paragraphs, the means of the BrW ratios in the various isoponderous groups are as follows: about 3.9 from tenrecs to prosimians, about 5.0 from tenrecs to marmosets, and about 1.7 from prosimians to monkeys. In log-log plots (Fig. 2), these ratios correspond to the distances between parallel lines. The lowest line is that through the data points of the tenrecs. A connecting line between the existing small tenrecs of the first three groups and the theoretically derived large tenrec of the 56-kg group has a slope of 0.624. Such a line predicts for each BoW an expected BrW of Tenrecinae. The BrWs of average Tenrecinae calculated from such a line are regarded to be uniformly 1, and BrW deviations in isoponderous species are expressed as encephalization indexes (EIs). The EIs calculated from this reference line closely correspond to those obtained from direct comparisons and two additional steps of adjustment (Fig. 1), and differ from them mainly by taking into account the relatively small differences in BoW existing within each of our five isoponderous groups.

The preceding comparison of BrWs in isoponderous species provides solid criteria for evaluating the slope of the best-fit line. Only a slope close to 0.624 produces EIs similar to those obtained from direct BrW comparisons (see below, Slopes Across Orders). Thus, our EIs are a reliable measure of RBrS.

Slopes Obtained From Interspecific Regression Analyses

Figures 1 and 2 show that species that are equidistant from Tenrecinae belong to the same family (Lorisidae, Callitrichidae) or suborder (Prosimii). Consequently, the lines connecting the data points of such taxonomic units must also have slopes close to 0.624. We found slopes of about 0.63 in genera, subfamilies, and families; about 0.68 in primate suborders; about 0.79 in the order Primates; and about 0.91 when the two orders (Insectivora and Primates) were combined. Since determining a common slope is crucial to our method of comparison, we present details of our material and procedure.

The reliability of the slope of a regression line depends on the range of BoWs. If the group under

investigation contains only species that are similar in BoW, the slope can be extremely high or low. In isoponderous species, the slope may even reach infinite values if the BrW differs only slightly. This is independent from the number of species. The slope is much more stable if the species in a group have a large range of BoW. Within genera, this range is generally limited [see Bronson, 1981].

Slopes Within Genera. In Insectivora we have eight, and in Primates 13 genera with more than one species. The average of the eight slopes of Insectivora is 0.575 (coefficient of variation, CV, is 63.9), and that of the 13 genera of Primates is 0.421 (CV 67.3). When considering only those genera in which the BoW of the largest species is at least twice that of the smallest (ie, BoW range >2), we obtain different results. Within the eight genera of Insectivora, four have a BoW range >2 . The average slope of these four genera, which all belong to the Soricidae, is 0.637 (CV 1.2). Among Primates, we have only three genera with a BoW range >2 ; their average slope is 0.577 (CV 44.5). Finally, the average of all the genera of Insectivora and Primates with a BoW range >2 is 0.611 (CV 24.8; $n = 7$). Since this is weak material and the CVs are relatively high, we will also include our large sample of Chiroptera containing 38 genera with more than one (up to 16) species: the average of the 38 slopes is 0.699 (CV 42.9). Among the 38 genera, there are 20 with a BoW range >2 (up to 1:8). The average of these 20 slopes is 0.633, and the variation is much smaller (CV 16.6). Our 12 species of the genus *Rhinolophus* are plotted in Figure 3. The average slope of well-represented genera is similar to that found from the direct comparison (Fig. 2).

Slopes Within Subfamilies and Families. All the regression lines of subfamilies and families with their respective BoW ranges are presented as full lines in Figure 4. For the four families of Insectivora that are well represented in our material, the following slopes were found (number of species in parentheses): Tenrecidae 0.562 (11), Erinaceidae 0.456 (3), Soricidae 0.612 (16), and Talpidae 0.661 (4). Within the Tenrecidae, the Oryzictinae and Potamogalinae have nearly twice the BrWs of Tenrecinae of similar BoWs. The slopes are 0.613 for Tenrecinae (4), 0.736 for Oryzictinae (4), and 0.689 for Potamogalinae (3). In shrews, the Soricidae have somewhat larger BrWs than Crociduri-

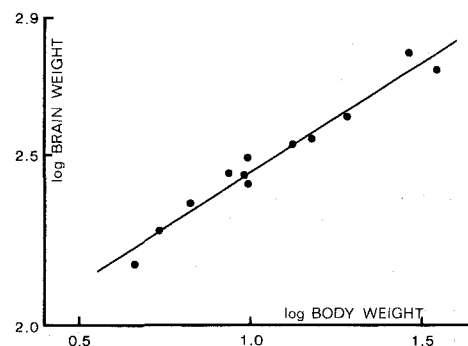


Fig. 3. Least-squares regression for the genus *Rhinolophus* to demonstrate the relations within a genus well represented both by number of species ($n = 12$) and range of BoW (1:7.6). The slope is 0.643; the correlation coefficient (r^2) is 0.958.

nae of similar BoWs [Fons et al, 1984]. The slope of Crocidurinae is 0.631 (9), and that of Soricinae is 0.680 (7). The average of the seven slopes of families or subfamilies of Insectivora is 0.638 (CV 14.1).

Two of the six families of prosimians have only one (Daubentoniidae) or few species (Tarsiidae). In the other four families, the following slopes were found (number of species in parentheses): Cheirogaleidae 0.625 (3), Lemuridae 1.103 (3), Indriidae 0.746 (4), Lorisidae 0.542 (6). In the Lemuridae, *Lepilemur* differs greatly from the rest of the family by its low BrW. If *Lepilemur* is excluded, the slope of Lemurinae [Rumpler, 1975] drops to 0.396 (2). The BoW range of this group, however, is only 1:2.1. The average of the slopes of the families (or subfamilies) of prosimians is 0.577 (CV 25.5; $n = 4$) including Lemurinae, and 0.638 (CV 16.1; $n = 3$) excluding them.

Of the five families of simians, one (Hominidae) is monospecific. In the remaining four families, the following slopes were found (number of species in parentheses): Callitrichidae 0.723 (7), Cebidae 0.658 (11), Cercopithecidae 0.525 (23), Pongidae 0.607 (7). In the Cercopithecidae there are two distinct subfamilies (Cercopithecinae and Colobinae) and one strongly deviating species (*Miopithecus*) whose small BoW has been attributed to dwarfism [Bauchot and Stephan, 1969; Frahm et al, 1982]. While Colobinae have distinctly lower EIs than Cercopithecinae, *Miopithecus* has a distinctly higher EI (Table 1). The slope of Cercopithecinae (*Miopithecus* excluded) is 0.613 (17); that of Colobinae is 0.467 (5). The average for the five

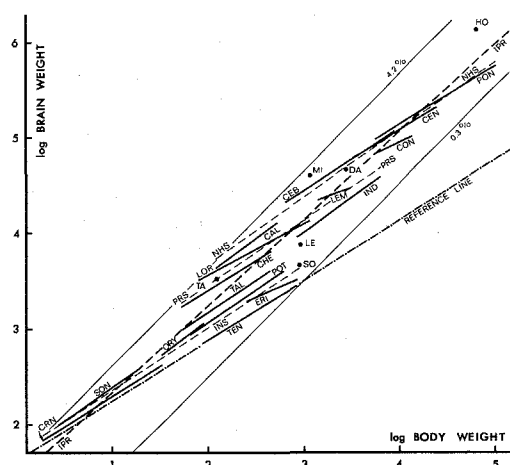


Fig. 4. Least-squares regression analyses of taxonomic units in Insectivora and Primates. Thick stippled line (IPR) is across the orders of Insectivora and Primates, thin stippled lines (INS, NHS, PRS) are from orders and suborders, and solid lines are from families and subfamilies. The slope of the reference line was determined from regression analyses within genera, subfamilies, and families. The reference line itself is placed through the average of the Tenrecinae, where the lowest RBrS was found. The equation for the reference line is $\log \text{BrW} = 1.6128 + 0.63 \times \log \text{BoW}$. Fine solid lines mark the limits of the percentage BrW relative to BoW in extant species of Insectivora and Primates (min = 0.3%, max = 4.2%, respectively). These lines are isometric (slope = 1). Abbreviations (single species have only two capitals; number of species in parentheses): CAL, Callitrichidae (7); CEB, Cebidae (11); CEN, Cercopithecinae (17); CHE, Cheirogaleidae (3); CON, Colobinae (5); CRN, Crocidurinae (9); DA, *Daubentonia*; ERI, Erinaceidae (3); HO, *Homo*; IND, Indriidae (4); INS, Insectivora (37); IPR, Insectivora and Primates (104); LE, *Lepilemur*; LEM, Lemurinae (2); LOR, Lorisidae (6); MI, *Miopithecus*; NHS, nonhuman simians (48); ORY, Oryzoricinae (4); PON, Pongidae (7); POT, Potamogalinae (3); PRS, prosimians (18); SO, *Solenodon*; SON, Soricinae (7); TA, *Tarsius* spp.; TAL, Talpidae (4); TEN, Tenrecinae (4).

slopes of families or subfamilies of simians is 0.614 (CV 15.4), and that for all families and subfamilies of Primates (Lemurinae excluded) is 0.623 (CV 14.6; $n = 8$). Finally, the average slope for all the families or subfamilies of Insectivora and Primates (Lemurinae excluded) is 0.630 (CV 13.9; $n = 15$).

A similar average slope (0.611) was obtained within genera (see above). Thus, there is no substantial difference between genera, subfamilies, and/or families in the slopes of the regression lines.

Slopes Within Suborders and Orders. In calculating the regression line for all the 37 species of Insectivora (fine broken line INS in Fig. 4), the common slope was found to be 0.648, which is only slightly higher than that found 1) for the genera (0.637) and 2) for subfamilies and families (0.638) (see above).

In the order Primates, the differences were much larger. The common slope was 0.792 including man ($n = 67$) and 0.773 excluding man. For the 18 prosimian species (fine broken line PRS in Fig. 4), we found a slope of 0.659, and for the 48 nonhuman simian species (fine broken line NHS in Fig. 4), the slope was 0.704. These slopes, too, are distinctly higher than those found from the genera, subfamilies, and/or families.

Slopes calculated from orders may be similar to those calculated from genera to families when species with different EIs are distributed over the whole BoW range of the order. This is the case in Insectivora, where the Talpidae, with high EIs, are counterbalanced by the Tenrecinae, with low EIs. No such balance exists in the Primates. Prosimians and Callitrichidae with low EIs are generally small-bodied (between 54 g and 6 kg), whereas monkeys, apes, and man, which have higher EIs, range from 0.6 to more than 100 kg. Very large prosimians, which may have reached gorilla size, are now extinct. The lack of small primates with high EIs and large primates with low EIs results in a regression line for the primate order, with a distinctly higher slope than the regression lines of lower taxonomic units (genera to families).

Slopes Across Orders. The common regression line of all the 104 species of Insectivora and Primates (including man) (thick broken line IPR in Fig. 4) has a slope of 0.907, which again is distinctly higher than all the slopes resulting from the lower taxonomic units. The main reason for this is the difference in the size range between Insectivora and Primates. Using the regression line across the two orders as a basis for reference, man would have a lower EI than many of the smaller primates (eg, *Miopithecus*). In the IPR line, the BoW power is too high, and thus the BoW influence on the EIs would not be minimized. The EIs would be dissim-

ilar in closely related groups and thus in contradiction with the direct comparisons (Fig. 1). The smaller species would have distinctly large EIs than the larger species. Both in Tenrecinae and in Prosimii, the EIs from the representatives of the 115-g group would be twice as large as those from the 850-g group. Thus a reference baseline across orders does not meet the requirement of producing similar RBrS values (expressed as EIs) for closely related species independent of differences in BoW.

Slope Selected for Interspecific Comparisons. Summarizing our investigation on slopes, we found an average of 0.611 for seven genera of Insectivora and Primates, 0.633 for 20 genera of bats, and 0.630 for 15 subfamilies and families of Insectivora and Primates. The values converge at 0.63, and this value will be used in the interspecific comparison of RBrS. Using this slope results in EIs similar to those found in direct BrW comparisons in isoponderous species and thus meets the fundamental requirement for employing the regression method to estimate actual species-specific differences in RBrS.

Variability of Slopes and Possible Explanations. In addition to the differences due to different taxonomic levels of the samples, there are several other possible reasons for the great variability of the slopes.

Poor Data. In a given species, the BoW data are generally more variable than the BrW data. Young animals generally have higher EIs. The reason for this is that BrW increases more rapidly than BoW to reach or even exceed adult values, and then remains nearly unchanged or decreases slightly. On the other hand, gravid females and overweight animals (eg, from zoos) have lower EIs. To minimize the influence of such specimens, we introduced for each species a standardized BoW obtained from as many specimens as possible, excluding obvious juveniles and very fat or very thin as well as sick animals, and by subtracting the uterus weight in gravid females. Despite these precautions, our standardized BoWs may differ markedly from the actual average BoW of a given species when the sample is poor and/or mainly based on data from the literature and/or on material from zoos. For example, our adult chimpanzees from the wild uniformly had BoWs close to 46 kg, whereas values of up to 80 kg are reported in the literature [Bauchot

and Stephan, 1969]. Our data come mainly from free-living animals, but there is no guarantee that they do not include poor specimens.

Number of species and body weight range. The range of BoW within a taxon is often too limited to arrive at stable slopes. This is especially true in primates, and of course poses more of a problem in lower taxonomic units than in higher ones. By using taxa with a great number of species, some of which are distinctly different in BoW, the slopes become highly stable (ie, show little change when new species are added), and they clearly converge on a common value for all genera and families and/or subfamilies (eg, in bats [Stephan et al, 1981b]).

Deviating species. In bats we have found (beside great uniformity) single, highly deviant species (eg, the giant form *Hipposideros commersoni* [Stephan et al, 1981b]). Such deviating species can be easily recognized in large samples. This is much more difficult in primates, since many primate families comprise only a few extant species. Furthermore, many, or even most, of the prosimian, cebid, and pongid species appear to be unique, and only a few genera (*Saguinus*, *Macaca*, *Cercopithecus*, *Presbytis*) comprise more than ten species. In taxa with few species, unique and/or deviating species will have a relatively great influence on the slope. If they are included in the regression line calculation, the fact that they are deviating forms may go unnoticed. Therefore, instead of considering the slopes obtained only from the weak material of primates, those obtained from related mammalian orders with large numbers of species, such as Insectivora and Chiroptera, should be taken into account when establishing the reference line for calculating EIs of primates. It is surprising that, despite the weak data on primates, the slopes of the various families and/or subfamilies are still quite similar (Fig. 4).

Brain Weight Comparison Based on Parallel Reference Lines

The following comparison includes only those species from which both BrWs and volumes of brain parts are known. In prosimians, the number of species is the same, but in nonhuman simians, volume measurements of brain parts, and thus calculation of size indexes (SIs, corresponding to the EIs of the total brain), were possible in only 26 of the 48 species used in Figure 4.

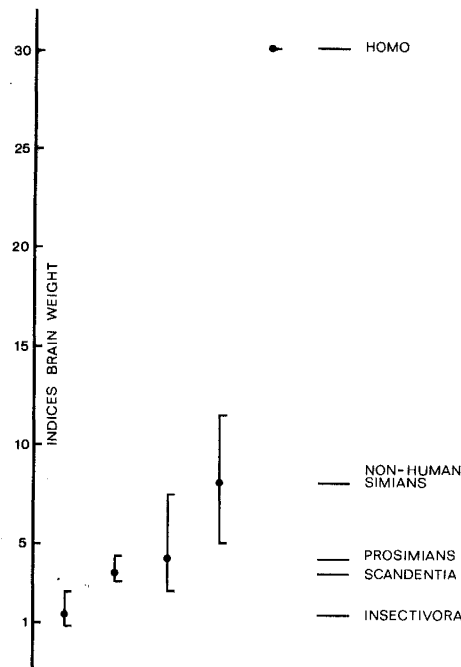


Fig. 5. Ranges and averages of the EIs of the large taxonomic units. The EIs are numerical values of distances from the reference line given in Figure 4. The reference line represents the "average tenrec," and the indexes of all points on it are equal to 1.

According to the results given above, the most meaningful slope is 0.63. For universal comparisons among all the primate groups, we have assumed this slope to be common to all the groups under comparison, even if in single groups the slopes resulting from the available data differ.

A reference line with a slope of 0.63 placed through the average of the Tenrecinae ($n = 4$) has a y intercept of 1.6128. Based on this line (ie, referring to Tenrecinae = 1), the mean EI is 1.38 for Insectivora, 3.48 for Scandentia, 4.15 for prosimians, 8.01 for nonhuman simians, and 30.12 for man (Fig. 5; Tables 1, 4). When the reference line is shifted to the prosimian average (ie, referring to the prosimian average = 1), the y intercept increases by 0.6178 ($= \log 4.15$) to 2.2306, and when it is displaced to the average of nonhuman simians (ie, referring to the nonhuman simian average = 1), the y intercept increases by 0.9035 ($= \log 8.01$) to 2.5163 (Table 2). All the indexes for the species given in Table 1 can be easily converted in this way (ie, by dividing them by

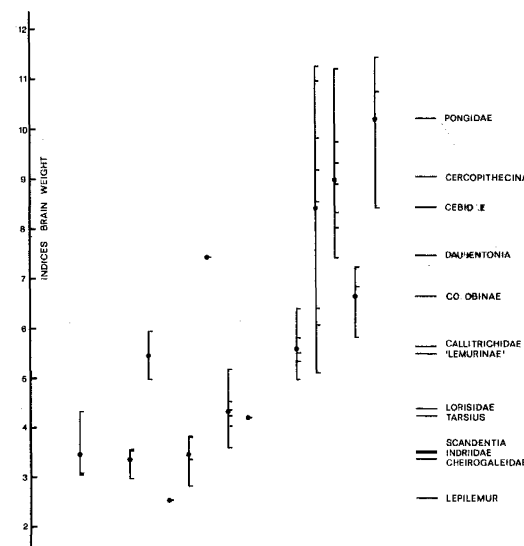


Fig. 6. EIs of Scandentia and lower taxonomic units of the 44 species of nonhuman primates (listed in Table 1) whose structures were measured. The vertical bars represent the range of variation within each taxonomic unit with the average marked by a dot. The horizontal bars to the right correspond with the means of the units, whose names are given.

4.15 or 8.01) to the prosimian or nonhuman simian reference baselines. Such conversion by constant factors corresponds to the subtraction (or in the reversed direction, to the addition) of constant y intercepts and thus to a parallel shifting of the reference line.

The EIs of Insectivora slightly overlap those of prosimians but not those of Scandentia (Fig. 5). The EIs of Scandentia completely overlap those of prosimians, but the average is distinctly lower. There is a wide overlap of the EIs of prosimians and simians, but all the prosimian EIs are below the nonhuman simian average. Man has an extremely high EI, and the distance between chimpanzee and man is larger than that between chimpanzee and Tenrecinae.

The EIs of prosimians have a wide range, which is largely due to the very high value of *Daubentonia* (Fig. 6). The lowest EI is found in *Lepilemur*. Within nonhuman simians, low EIs are found in Callitrichidae and Colobinae. Most of the Cebidae, together with Cercopithecinae and Pongidae, have higher EIs, which almost completely overlap in their ranges of variation. The differences in RBrS (expressed as EIs) will be discussed

in more detail in conjunction with differences in the relative size of the brain structures (expressed as SIs).

Summary

Comparison of brain weights in groups of isoponderous species gives reliable data on relative brain size, which can be used directly as criteria for validating the results obtained from regression analyses of species with different BoWs. RBrS values similar to those obtained from isoponderous groups result only when using BoW powers (slopes a in log-log plots) slightly higher than 0.6. Such slopes are obtained in regression analyses for genera, subfamilies, and families but not for species (lower values), suborders, or orders or across mammalian orders (generally higher values). Within the taxonomic units from genera to families, the slopes may show a great variability. This seems mainly to be due to poor material, since in samples with large numbers of species having a sufficient range of BoWs the slopes become more stable and converge at 0.63. A reference line with this slope was placed through the average of Tenrecinae, which in all our isoponderous groups (which include Insectivora; Fig. 1) had the smallest brains. The equation of this line is $\log BrW = 1.6128 + 0.63 \times \log BoW$. Based on this line, prosimians have an average brain size enlargement (EI) of 4.15, nonhuman simians of 8.01, and man of 30.12. These EIs are independent of differences in BoW. The EIs of prosimians partly overlap those of simians; in both groups, the values diverge enormously.

COMPARISON OF BRAIN STRUCTURES

The methods and procedures for size comparisons of brain structures are the same as those used for comparisons of total BrW. Using photoprints of serial sections, the total brain was subdivided into the five basic brain parts (medulla OB-Longata, MESencephalon, CERebellum, DIEncephalon, and TELencephalon), and the TEL was subdivided into seven parts (Bulbus OLfactorius, PALeocortex, SEPTum, HIPpocampus, SCHizo-cortex, STRiatum, and NEOcortex). The borders of the brain parts were delineated on the photoprints, and the components were cut out and weighed. The weights were converted to the fresh

TABLE 1. Body Weights (BoWs in g), Brain Weights (BrWs in mg), and Encephalization Indexes (EIs) of Tenrecinae, Scandentia, and the 45 Primate Species Whose Brain Structures Were Measured. Some Ecoethological Characteristics Are Given for the Primate Species

	BoW	BrW		Ecoethological characteristics ^b			
	(g)	(mg)	EI ^a	(1)	(2)	(3)	(4)
Tenrecinae							
0577 <i>Tenrec ecaudatus</i>	832	2,570	0.91				
0579 <i>Setifer setosus</i>	248	1,510	1.14				
0581 <i>Hemicentetes semispinosus</i>	110	830	1.05				
0585 <i>Echinops telfairi</i>	87.5	620	0.90				
Scandentia							
1263 <i>Tupaia glis</i>	170	3,200	3.07				
1269 <i>Tupaia minor</i>	120	2,580	3.08				
1289 <i>Urogale everetti</i>	275	4,280	3.03				
Prosimians							
3199 <i>Microcebus murinus</i>	54	1,780	3.52	N	1	AQP	4,8,13,16,20,21
3203 <i>Cheirogaleus major</i>	450	6,800	3.53	N	2	AQP	4,8,10,13,20,21
3205 <i>Cheirogaleus medius</i>	177	3,140	2.94	N	2	AQP	4,8,13,16,21
3216 <i>Lemur fulvus</i>	1,400	23,300	5.92	D	3a	AQP	8,10,16,20,21
3227 <i>Varecia variegata</i>	3,000	31,500	4.95	D	3a	AQP	4,7,8,20,21
3239 <i>Lepilemur ruficaudatus</i>	915	7,600	2.53	N	3b	VCL	4,5,16,21
3243 <i>Avahi laniger laniger</i>	1,285	10,490	2.81	N	3b	VCL	4,5,8,10,19,20,21
3244 <i>Avahi l. occidentalis</i>	860	9,670	3.34	N	3b	VCL	4,5,8,10,19,20,21
3247 <i>Propithecus verreauxi</i>	3,480	26,700	3.82	D	3c	VCL	4,5,8,10,16,17,20,21
3249 <i>Indri indri</i>	6,250	38,300	3.79	D	3c	VCL	4,5,7,10,16,17,20
3251 <i>Daubentonia mada-gascariensis</i>	2,800	45,150	7.42	N	1	AQP	4,8,16,20,21
3253 <i>Loris tardigradus</i>	322	6,600	4.23	N	1	SCQ	4,7,8,10,16,21
3255 <i>Nycticebus coucang</i>	800	12,500	4.52	N	1	SCQ	4,7,8,16,21
3259 <i>Perodicticus potto</i>	1,150	14,000	4.03	N	2	SCQ	4,7,8,10,13,15,16,21
3265 <i>Galago senegalensis</i>	186	4,800	4.35	N	2	VCL	4,8,15,16,21
3267 <i>Otolemur crassicaudatus</i>	850	10,300	3.59	N	2	VCL	4,7,10,13,15,16
3275 <i>Galagoides demidoff</i>	81	3,380	5.17	N	1	VCL	4,8,10,15,16,21
3282 <i>Tarsius spp.</i>	125	3,600	4.19	N	1	VCL	4,7,8,18,21

^aEIs are based on the Tenrecinae reference line (slope 0.63; y intercept 1.6128).

^bKey: (1) D, diurnal; N, nocturnal. (2) 1, insectivores; 2, frugivores; 3, folivores; 3a, generalist folivores; 3b, nocturnal strict folivores; 3c, diurnal strict folivores. (3) AQP, arboreal quadrupedal prosimians; NAQ, New World arboreal quadrupedal simians; NSB, New World semibrachiators; OAQ, Old World arboreal quadrupedal simians; OMB, Old World modified brachiators; OSB, Old World semibrachiators; OTB, Old World vertical brachiators; OTQ, Old World terrestrial quadrupedal simians; SBP, specialized bipedal hominid; SCQ, slow-climbing quadrupedal prosimians; VCL, vertical clinging and leaping prosimians. (4) References: 1, Erikson [1963]; 2, Napier [1963]; 3, Knussmann [1967]; 4, Napier and Napier [1967]; 5, Hill [1972]; 6, Christen [1974]; 7, Roberts [1974]; 8, Walker [1974]; 9, Hladik [1975]; 10, Kay [1975]; 11, Clutton-Brock and Harvey [1977]; 12, Hershkovitz [1977]; 13, Seligsohn [1977]; 14, Kay [1978]; 15, Charles-Dominique and Bearder [1979]; 16, Hladik [1979]; 17, Klopfer and Boskoff [1979]; 18, Niemitz [1979]; 19, Petter and Charles-Dominique [1979]; 20, Pollock [1979]; 21, Walker [1979].

TABLE 2. Slopes and y Intercepts of Reference Lines, and Conversion Factors for Indexes

	Slope (a)	y Intercepts			Conversion factors ^a		
		Tenrecinae	Prosimians	Nonhuman simians	Tenr/ pros	Tenr/ sim	Pros/ sim
Total brain	0.63	1.6128	2.2306	2.5163	4.15	8.01	1.93
Medulla oblongata	0.60	0.7930	0.9881	1.0550	1.57	1.83	1.17
Mesencephalon	0.54	0.5690	1.0079	1.1052	2.75	3.44	1.25
Cerebellum	0.67	0.5690	1.2422	1.3890	4.71	6.61	1.40
Diencephalon	0.60	0.4765	1.2111	1.3731	5.43	7.88	1.45
Telencephalon	0.63	1.3570	2.0436	2.3845	4.86	10.65	2.19
Bulbus olfactorius	0.57	0.8440	0.5670	-0.2385	0.53	0.08	0.15
Paleocortex	0.57	1.0391	0.9680	0.9212	0.85	0.76	0.90
Septum	0.57	-0.0346	0.2383	0.2715	1.87	2.02	1.08
Hippocampus	0.57	0.5870	1.0438	1.0100	2.87	2.65	0.92
Schizocortex	0.60	0.0805	0.6234	0.5508	3.49	2.95	0.85
Striatum	0.60	0.2570	1.0244	1.2540	5.85	9.93	1.70
Neocortex	0.67	0.4790	1.7944	2.1836	20.65	50.62	2.45
Area striata gray	0.51	—	1.2980	1.6664	—	—	2.34
Corpus geniculatum laterale	0.48	-1.2699	0.0908	0.3514	22.95	41.81	1.82

^aTo convert the indexes given in Tables 1 and 3-6 (Tenrecinae basis) to prosimian or simian bases, they must be divided by the factors given in the first two columns. The factors in the third column are for conversions between prosimians and simians.

volume of the standardized brain of a given species. The volume data as well as details of all the steps of measurement and conversion are published elsewhere [Stephan et al, 1981a]. The same methods as for the total brain were used to determine common slopes for each of the brain parts. When the brain components were plotted against BoW, the slopes were different for the various brain parts. They vary from 0.54 for the MES to 0.67 for CER and NEO (Table 2). Slopes lower than 0.54 have been found in some visual structures—eg, 0.48 for the lateral geniculate body [Stephan et al, 1984] and 0.51 for the gray matter of the area striata [Frahm et al, 1984]; a distinctly higher slope than 0.67 has been found for the white matter of the NEO (0.86) [Frahm et al, 1982].

The SIs were scaled for detailed comparisons between and within the various taxonomic units. The reference base groups were 1) the Tenrecinae for comparison of the orders (Insectivora, Scandentia), primate suborders (prosimians, simians excluding man) and for strongly deviating species (man) (Figs. 7-9); 2) the primate suborders for families and subfamilies and for species that markedly deviate from these taxonomic units (Figs. 10, 11); and 3) New World families for their species (Fig. 12). As mentioned above, conversion from one reference basis to another is easy using the constant factors given in Table 2.

To obtain a general indication of the trends (whether progressive or regressive) and intensities of the size modifications within the brains, the SIs were averaged (basis Tenrecinae; Table 4), and the mean SIs of Insectivora (Tenrecinae excluded), Scandentia, prosimians, nonhuman simians, and *Homo* were scaled. In Figures 7-9 the Scandentia were positioned between Insectivora and prosimians with no regard for their classification.

Of the five fundamental brain parts, the smallest increase in the SIs was found for OBL (Fig. 7; Tables 3, 4). The mean SI in Scandentia is slightly higher than in prosimians. A stronger increase was found for the MES. The mean SI of the midbrain is very high in Scandentia, and higher than that of prosimians and simians. The mean SIs of the remaining three parts (CER, DIE, TEL) are even higher and varied in height and sequence. In prosimians, the mean SIs are lower than in simians, and the DIE has the highest value, whereas in simians the mean of the TEL is the highest. In nonhuman simians, the mean SI of DIE is higher

than that of CER, whereas in man the sequence is reversed. Overall, there is a steady increase of the SIs of all basic brain parts from Tenrecinae through other Insectivora, prosimians, and nonhuman simians up to man, whereas Scandentia deviated from these trends with a relatively large DIE and an especially large MES (Fig. 7).

Within the TEL (Fig. 8), the mean SIs of the BOL become distinctly lower, and those of the PAL (amygdala included) become slightly lower (except in Scandentia and man). Whereas in the limbic structures (SEP, HIP, and SCH) the increase of the mean SIs is modest, it is strong in STR and very strong in NEO. In SCH, SEP, PAL, and BOL, Scandentia again do not follow the trend, having relatively high mean SIs in all these structures. In SCH and HIP, the nonhuman simians have lower mean SIs on the average than the prosimians. Thus, these limbic structures show no uniform trend.

The NEO is by far the most progressive structure. In Insectivora (Tenrecinae excluded), the NEO is on the average 2.7 times larger than in Tenrecinae; in prosimians, 21 times; in nonhuman simians, 51 times; and in man, 200 times (Fig. 9). The Scandentia, when positioned between Insectivora and prosimians, are in line with this general trend.

In summary, the slopes of the various brain structures range from 0.48 (lateral geniculate body) to 0.86 (white matter of the neocortex). Thus, changes in body size do not affect size modifications of all brain parts at the same rate.

A classification of the various brain components based on the trends and intensities of size modifications from low Insectivora through prosimians, monkeys, and apes to man may be suggested as follows: an extremely strong progressive structure is the neocortex (NEO); strongly progressive structures are striatum (STR), cerebellum (CER), and diencephalon (DIE); fairly progressive structures are schizocortex (SCH), septum (SEP), hippocampus (HIP), and mesencephalon (MES); conservative structures are medulla oblongata (OBL) and paleocortex + amygdala (PAL); and regressive structures are olfactory bulb (BOL) and PAL. The PAL is regressive in nonhuman primates but conservative in man. This may be because the olfactory components of this complex decrease, whereas, especially in man, the limbic components (amygdaloid complex) markedly increase [Stephan and Andy, 1977].

TABLE 3. Indexes of Brain Structures Based on the Tenrecinae Reference Line of Each Structure

	n(i)	OBL	MES	CER	DIE	TEL	BOL	PAL	SEP	HIP	SCH	STR	NEO
Tenrecinae													
0577 <i>Tenrec ecaudatus</i>	3	0.89	0.95	0.92	0.90	0.89	0.98	0.92	0.86	0.82	1.13	0.96	0.99
0579 <i>Setifer setosus</i>	2	1.08	1.04	0.95	1.22	1.23	1.30	1.33	1.22	1.11	1.27	1.15	1.11
0581 <i>Hemicentetes semispinosus</i>	1	1.04	0.95	1.35	0.97	1.00	0.92	0.92	0.95	1.13	0.86	0.98	1.05
0585 <i>Echinops telfairi</i>	2	0.99	1.06	0.78	0.91	0.88	0.80	0.83	0.97	0.94	0.74	0.91	0.85
Scandentia													
1263 <i>Tupaia glis</i>	2	1.73	4.46	3.56	4.44	3.12	0.99	1.27	1.89	1.94	3.26	3.46	10.9
1269 <i>Tupaia minor</i>	2	1.58	5.06	3.36	4.75	3.12	0.90	1.22	1.84	2.23	4.32	3.88	10.4
1289 <i>Urogale everetti</i>	1	1.97	3.99	3.21	4.52	3.10	1.09	1.34	1.99	1.82	3.55	3.79	10.3
Prosimians													
3199 <i>Microcebus murinus</i>	6	1.46	2.67	4.36	4.07	4.02	0.66	0.97	1.70	2.67	3.04	4.33	17.0
3203 <i>Cheirogaleus major</i>	2	1.57	2.40	4.26	4.18	4.04	0.71	0.93	1.96	2.76	3.17	4.63	16.3
3205 <i>Cheirogaleus medius</i>	2	1.46	2.39	3.68	3.52	3.27	0.78	0.95	1.84	2.36	2.43	3.59	12.6
3216 <i>Lemur albifrons fulvus</i>	3	1.90	3.18	7.00	7.41	7.13	0.48	0.94	2.07	3.13	4.55	8.71	31.6
3227 <i>Varecia variegata</i>	1	1.88	3.64	5.41	6.92	5.80	0.56	0.94	2.32	3.79	4.14	7.22	23.8
3239 <i>Lepilemur ruficaudatus</i>	3	1.21	1.77	3.26	3.31	2.82	0.34	0.57	1.26	2.08	2.70	3.38	11.3
3243 <i>Avahi laniger laniger</i>	2	1.22	1.94	3.32	3.92	3.17	0.22	0.44	1.28	2.30	2.73	3.96	13.2
3244 <i>Avahi l. occidentalis</i>	2	1.42	2.42	4.03	4.99	3.75	0.22	0.52	1.59	2.62	3.20	4.59	15.9
3247 <i>Propithecus verreauxi</i>	2	1.48	2.57	4.52	5.50	4.40	0.21	0.58	1.65	2.59	2.76	5.85	18.5
3249 <i>Indri indri</i>	2	1.14	2.35	4.25	5.13	4.56	0.17	0.53	1.52	2.70	3.72	5.42	19.1
3251 <i>Daubentonia madagascariensis</i>	1	2.09	3.33	8.54	10.1	8.94	1.08	1.52	2.98	4.98	7.41	13.1	36.0
3253 <i>Loris tardigradus</i>	1	1.17	2.62	4.10	5.62	5.26	0.47	0.87	1.65	1.84	2.59	6.07	24.4
3255 <i>Nycticebus coucang</i>	2	1.54	2.52	4.01	6.52	5.54	0.52	1.03	2.21	3.24	3.19	7.62	23.3
3259 <i>Perodicticus potto</i>	2	1.60	2.35	4.08	4.98	4.88	0.81	1.09	2.22	2.83	3.97	5.74	19.7
3265 <i>Galago senegalensis</i>	1	1.78	3.30	5.47	5.55	4.90	0.60	0.76	1.97	3.43	3.87	5.05	21.4
3267 <i>Otolemur crassicaudatus</i>	2	1.52	2.71	4.16	4.58	4.11	0.52	0.75	1.62	2.55	2.63	5.37	17.1
3275 <i>Galagoides demidoff</i>	2	1.95	3.40	5.87	5.87	6.18	0.99	1.35	2.27	3.22	3.56	6.71	27.4
3282 <i>Tarsius</i> spp.	2	1.84	3.91	4.55	5.58	4.74	0.18	0.54	1.61	2.52	3.16	4.05	23.1
Simians													
3287 <i>Callithrix jacchus</i>	4	1.74	3.79	4.68	6.29	6.71	0.13	0.70	2.17	2.31	2.54	7.00	33.3
3289 <i>Cebuella pygmaea</i>	2	1.69	3.61	4.71	5.96	6.15	0.11	0.73	1.90	2.07	3.55	5.03	31.4
3305 <i>Saguinus midas</i>	2	2.09	3.85	5.76	6.99	7.88	0.09	0.80	1.60	2.62	3.02	7.88	39.3
3311 <i>Saguinus oedipus</i>	3	1.89	3.64	4.96	7.13	7.35	0.09	0.79	2.22	2.30	2.51	7.11	36.6
3315 <i>Callimico goeldii</i>	1	1.82	3.27	5.34	6.07	6.95	0.12	0.68	2.04	2.15	2.83	6.72	34.3
3317 <i>Cebus albifrons</i>	2	2.25	4.29	9.72	10.7	14.5	0.06	0.87	1.93	2.36	2.60	14.5	70.6
3325 <i>Aotus trivirgatus</i>	5	1.96	2.93	5.60	6.50	7.72	0.17	0.78	1.97	3.02	3.59	8.45	36.6
3327 <i>Callicebus moloch</i>	2	2.14	3.63	4.70	7.75	8.15	0.06	0.86	1.94	3.15	3.29	8.59	38.9
3335 <i>Saimiri sciureus</i>	1	2.36	4.26	7.87	9.71	13.0	0.10	0.93	2.42	2.25	2.84	11.7	66.6
3341 <i>Pithecia monachus</i>	2	2.02	3.87	7.85	9.48	10.9	0.08	0.95	2.34	3.34	2.99	13.2	52.0
3363 <i>Alouatta seniculus</i>	2	1.34	2.34	4.33	5.81	6.57	0.04	0.51	1.46	2.31	2.21	8.15	29.6
3371 <i>Ateles geoffroyi</i>	1	1.34	3.12	8.14	8.10	12.2	0.08	0.89	2.09	2.11	2.77	12.5	57.1
3379 <i>Lagothrix lagothricha</i>	3	2.00	4.20	9.84	11.3	15.0	0.08	0.97	2.20	3.13	3.33	16.1	70.8
3391 <i>Macaca mulatta</i>	1	1.48	2.94	5.97	6.91	11.0	0.07	0.67	1.77	2.12	2.45	10.3	52.0
3407 <i>Cercopithecus albigena</i>	1	2.00	3.75	7.08	8.19	11.9	0.10	0.90	1.91	2.31	2.40	10.5	55.8
3415 <i>Papio anubis</i>	2	1.96	3.08	5.70	7.12	11.6	0.13	0.60	1.88	2.74	2.50	9.13	52.6
3433 <i>Cercopithecus ascanius</i>	1	2.00	3.88	6.77	9.15	13.4	0.14	0.93	2.64	2.99	4.38	11.9	64.5
3453 <i>Cercopithecus mitis</i>	1	1.69	3.24	5.19	7.32	9.99	0.11	0.79	1.82	2.42	2.69	7.94	47.2
3469 <i>Miopithecus talapoin</i>	2	2.37	4.84	7.87	11.3	15.2	0.07	1.13	2.53	3.21	3.06	15.0	75.9
3473 <i>Erythrocebus patas</i>	2	1.95	3.46	5.82	8.37	13.2	0.04	0.74	2.16	2.49	2.66	9.27	63.2
3477 <i>Colobus badius</i>	2	1.59	3.02	6.19	6.49	9.62	0.05	0.55	2.01	2.78	3.33	8.78	44.8
3493 <i>Pygathrix nemaeus</i>	1	1.68	2.93	5.51	7.47	8.94	0.01	0.53	1.94	3.67	2.84	8.29	41.0
3499 <i>Nasalis larvatus</i>	1	1.54	2.42	5.45	5.77	7.61	0.02	0.50	1.56	2.20	2.31	6.72	34.7
3539 <i>Hylobates lar</i>	1	2.02	3.69	9.92	10.6	14.4	0.05	0.83	2.36	5.00	5.26	14.8	66.5
3549 <i>Pan troglodytes</i>	1	1.49	3.06	8.85	8.19	15.9	0.08	0.55	2.03	2.15	2.67	10.8	72.7
3551 <i>Gorilla gorilla</i>	1	1.17	2.28	8.08	6.28	11.2	0.06	0.61	1.74	1.70	2.20	7.83	49.0
3553 <i>Homo sapiens</i>	1	2.01	5.49	22.1	14.4	43.4	0.03	1.49	5.10	4.81	6.61	20.6	199.2

Abbreviations: BOL, olfactory bulb; CER, cerebellum; DIE, diencephalon; HIP, hippocampus; MES, mesencephalon; NEO, neocortex; OBL, medulla oblongata; PAL, paleocortex + amygdala; SCH, schizocortex; SEP, septum; STR, striatum; TEL, telencephalon; n(i) number of individuals.

TABLE 4. Average Indexes and Coefficients of Variation of Total Brains (TBR) and Brain Structures in Taxonomic Units (based on the Tenrecinae reference line)

	n(s)	TBR	OBL	MES	CER	DIE	TEL	BOL	PAL	SEP	HIP	SCH	STR	NEO
Scandentia	3	3.06	1.76	4.50	3.38	4.57	3.11	0.99	1.28	1.91	2.00	3.71	3.71	10.5
Prosimians	18	4.15	1.57	2.75	4.71	5.43	4.86	0.53	0.85	1.87	2.87	14.8	6.0	2.9
Cheirogaleidae	3	3.33	1.50	2.49	4.10	3.92	3.78	0.72	0.95	1.83	2.60	2.88	4.18	15.3
Lemurinae	2	5.44	1.89	3.41	6.21	7.17	6.47	0.52	0.94	2.20	3.46	4.35	7.97	15.2
Indridae	4	3.44	1.32	2.32	4.03	4.89	3.97	0.21	0.52	1.51	2.55	3.10	4.96	16.7
Lorisidae	6	4.32	1.59	2.82	4.62	5.52	5.15	0.65	0.98	1.99	2.85	3.30	6.09	22.2
Lorisinae	3	4.26	1.44	2.50	4.06	5.71	5.23	0.60	1.00	2.03	2.64	3.25	6.48	22.5
Galaginae	3	4.37	1.75	3.14	5.17	5.33	5.06	0.70	0.95	1.95	3.07	3.35	5.71	22.0
Nonhuman simians	26	8.01	1.83	3.44	6.61	7.88	10.7	0.08	0.76	2.02	2.65	2.95	9.93	50.6
Callitrichidae	5	5.59	1.85	3.63	5.09	6.49	7.01	0.11	0.74	1.99	2.29	2.89	6.75	28.7
Cebidae	8	8.39	1.93	3.58	7.26	8.67	11.0	0.08	0.85	2.04	2.71	2.95	11.6	35.0
Cercopithecidae	10	8.26	1.83	3.36	6.16	7.81	11.2	0.07	0.73	2.02	2.69	2.86	9.79	53.2
Cercopithecinae	7	8.98	1.92	3.60	6.34	8.33	12.3	0.09	0.82	2.10	2.61	2.88	10.6	58.7
Colobinae	3	6.59	1.60	2.79	5.72	6.58	8.72	0.03	0.53	1.84	2.88	2.83	7.93	16.6
Pongidae	3	10.2	1.56	3.01	8.95	8.37	13.8	0.06	0.66	2.04	2.95	3.38	11.1	62.7
		15.7	27.5	23.5	10.3	26.1	17.5	24.1	22.2	15.2	60.7	48.8	31.3	19.6

Abbreviations of structures as in Table 3; n(s), number of species.

INTERPRETATION OF SIZE DIFFERENCES IN BRAINS AND BRAIN PARTS

Characteristics of Taxonomical Groups

The large increase in RBrS, which is mainly due to an increase of the NEO, may be considered one of the prime factors in primate evolution and indicative of the successful progress of these mammals. Adaptive radiation, which has been very diverse in primates, is reflected by the differential growth of individual brain parts in the various groups.

Although they show many of the basic characters that might be expected in primitive primates, tree shrews (Scandentia) differ from prosimians by higher SIs of brainstem components and olfactory centers but lower SIs of CER, STR, and NEO. Either limbic structures have lower SIs (HIP) or the SIs are within the range of variation in prosimians. Visual components show a generally weak development with high adaptation to diurnal activities [Frahm et al, 1984].

Prosimians. Prosimians are characterized by a considerable variety of forms, which may be grouped into three infraorders: Lemuriformes, Lorisiformes and Tarsiiformes [Gregory, 1915]. Tarsiers are highly specialized, showing a number of adaptations, some of which suggest that they are more closely related to simians. Therefore, tarsiers and simians have been combined as Haplorhini [Pocock, 1918]. However, most of their brain features bear a greater resemblance to those of prosimians [Stephan, 1965, 1984; Bauchot, 1982], and we have therefore maintained the subdivision into prosimians and simians. The extant tarsiers are relatively uniform, whereas the Lemuriformes on Madagascar and the Lorisiformes in Africa and Southeast Asia have radiated to occupy a wide variety of niches. Because of the absence of simians on the island of Madagascar, the Lemuriformes could occupy niches that are elsewhere occupied by simians. Thus, Lemuriformes are more diversified than Lorisiformes. This adaptive diversity is equally reflected by the diversity in quantitative brain composition, but no clear trends in brain composition could be found that would permit a separation of the two groups.

In *Daubentonia* (Daubentoniidae), all brain structures with the exception of the MES have higher SIs than in any other prosimian (Fig. 10). The high SIs may be considered a consequence of secondary reduction in BoW (dwarfism) [Bauchot

and Stephan, 1969; Frahm et al, 1982]. *Daubentonia* differs in many ways from the other Lemuriformes. It has rodentlike incisor teeth, elongated digits, and a very long and slender third finger, which is used to dig insects from the bark of trees. According to Tattersall [1982], *Daubentonia* is a highly adaptable form found in numerous habitats but never in high density.

In Cheirogaleidae, the only structures with higher SIs than the average prosimian are BOL and PAL (Fig. 10). The SIs of the strongly progressive brain components, such as DIE and TEL but especially STR and NEO, show low mean values compared with other prosimian families. The external morphology of the brain shows conservative features too [see Stephan et al, 1977]. Other morphological and behavioral features are also characterized by retention of the primitive state [Hill, 1972].

Lemuridae comprise two subfamilies: Lemurinae and Lepilemurinae [Rumpler, 1975]. In Lemurinae, the mean SIs of all brain structures (except BOL) are higher than the mean SIs of the total sample of prosimians (Fig. 10). The strongly progressive structures (NEO, STR) reach particularly high SIs. With the exception of *Daubentonia*, *Lemur* has the highest SI of the NEO. The species of the genus *Lemur*, and particularly *Lemur fulvus*, are the most successful primates on Madagascar [see Petter and Petter-Rousseaux, 1979]. Despite a higher EI than the average prosimian, *Varecia* does not reach the EI of *Lemur*. *Varecia*, highly confined in its distribution, has a number of primitive characters that distinguish it markedly from *Lemur* [Petter and Petter-Rousseaux, 1979].

Lepilemur, which occupies a very different niche, differs dramatically in EI and brain composition from the Lemurinae. In many cases it has the lowest SIs of all prosimians (Fig. 10), and the SI of its NEO is only slightly higher than that of *Tupaia*. Differences in morphological and behavioral features gave rise to an increasingly stronger separation from other Lemuridae. *Lepilemur* was given tribe rank by Stephan and Bauchot [1965], subfamily rank by Rumpler [1975], and family rank by Petter and Petter-Rousseaux [1979]. In addition to its very distinctive karyotype [Rumpler, 1975], *Lepilemur* approaches closely what may be called ancestral lemurine morphology in its dental morphology [Seligsohn, 1977; Szalay and Delson, 1979].

In Indridae, all mean SIs are below the averages found for the prosimian sample as a whole (Fig.

10). In NEO and STR, the two diurnal forms (*Indri*, *Propithecus*) have distinctly higher mean SIs than the nocturnal *Avahi*. It is interesting to note that the *Avahi* has the most primitive karyotypes of the Indriidae [Petter and Petter-Rousseaux, 1979]. Unlike *Indri* and *Propithecus*, *Avahi* is a relatively solitary type, forming at the most family units of 2-4 individuals [Klopfer and Boskoff, 1979].

The two subfamilies of the Lorisidae (Loriformes) differ from each other by a series of morphological and behavioral characteristics [see Szalay and Delson, 1979] but show no distinctive trends in brain composition. From the brain components, the strongly progressive NEO, STR, and DIE have slightly higher mean SIs in Lorisiae than in Galaginae. On the other hand, brainstem components (OBL, MES) and BOL have higher mean SIs in Galaginae than in Lorisiae. Of the motor centers, the CER has a higher mean SI in Galaginae, and the STR has a lower one. The limbic structures also have higher mean indexes at times in Galaginae (HIP, SCH) and at times in Lorisiae (SEP, PAL).

Interspecific differences in the SIs are more marked in Galaginae than in Lorisiae (Table 4). In several cases (eg, NEO, BOL) *Galagoides* has the highest SI of the Lorisidae and *Otolemur* has the lowest. The SI of the NEO, which in *Galagoides* is even one of the highest in prosimians, seems to be inconsistent with the view that this form represents a conservative primate pattern [Eisenberg, 1981].

In *Tarsius*, the SIs of the brainstem components surpass the prosimian average. The SI of NEO is high for a prosimian; this is true even if the rela-

Figs. 7-9. Average size indexes (SIs; number of species in parentheses) of the brain structures in Tenrecinae (4), Insectivora except Tenrecinae (31), Scandentia (3), prosimians (18), nonhuman simians (26), and man. The average SI of each of the structures in the four Tenrecinae is uniformly given the value 1.

Fig. 7. The five fundamental brain components: ○, medulla oblongata (OBL); ■, mesencephalon (MES); ◇, cerebellum (CER); ▲, diencephalon (DIE); +, telencephalon (TEL).

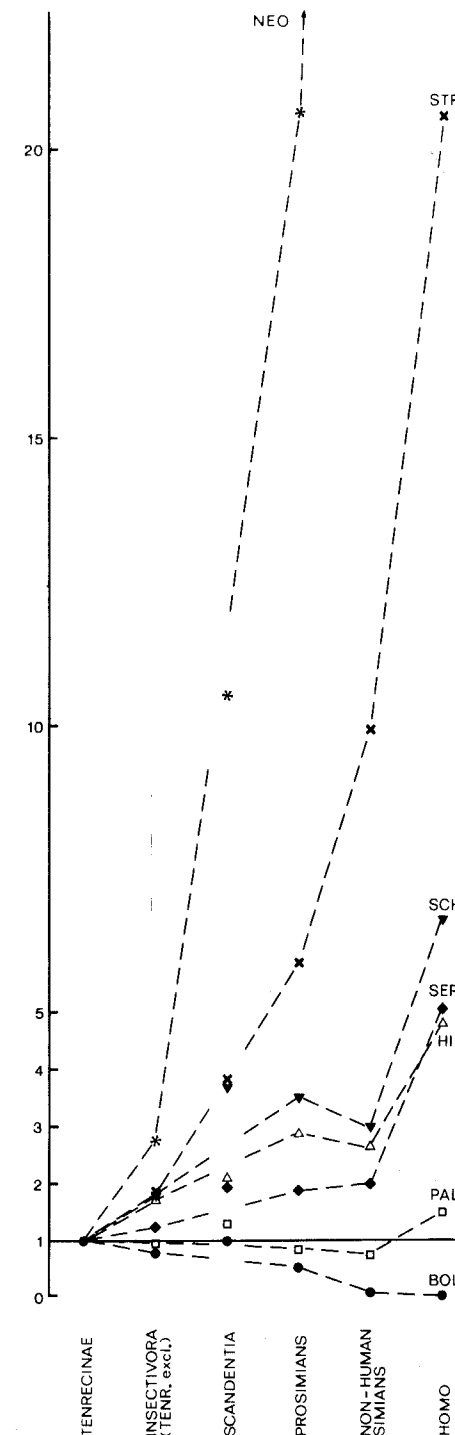
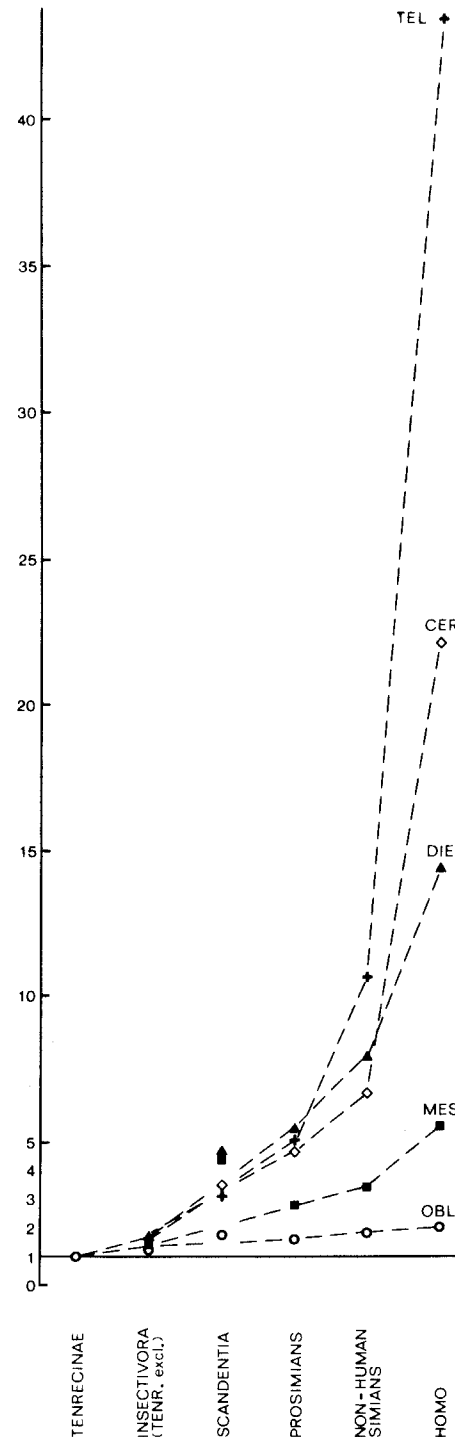


Fig. 8. The seven telencephalic components: ●, olfactory bulb (BOL); □, paleocortex + amygdala (PAL); ◆, septum (SEP); △, hippocampus (HIP); ▼, schizocortex (SCH); X, striatum (STR); *, neocortex (NEO).

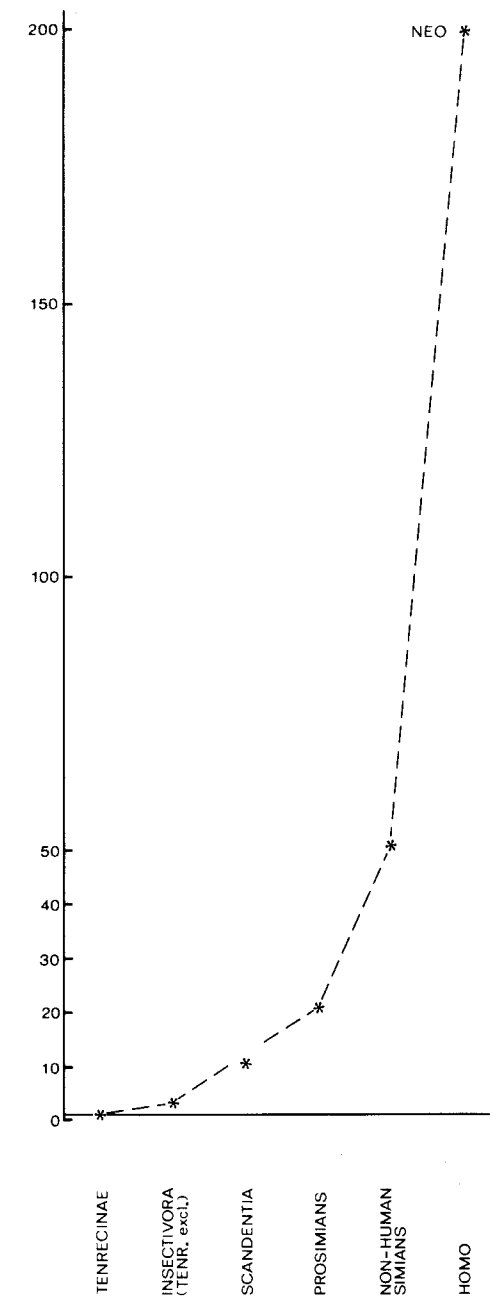


Fig. 9. The neocortex (*) in a reduced scale to show the position of nonhuman simians and man true to scale.

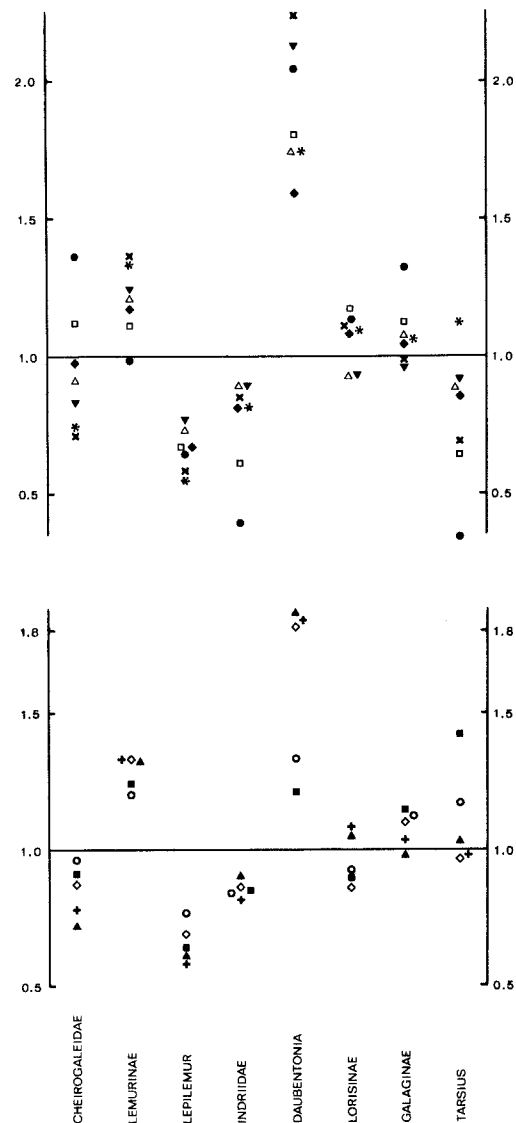


Fig. 10. Size indexes (partly means) of the brain components in prosimian taxonomic units. The mean SI of each component in the 18 prosimian species is uniformly set at 1. In the lower part, the five fundamental brain components; in the upper part, the seven telencephalic components. Symbols as in Figures 7-9.

tively large area striata is extracted from the NEO [Stephan, 1984]. The motor centers, such as STR and CER, have lower SIs than the prosimian average. Similarly lower SIs have been found for diencephalic motor structures [Baron, 1979]. The SIs of

the olfactory structures are also far below the prosimian average; those of the limbic structures are near, but still below the means for prosimians (Fig. 10).

Simians. The simians (New World and Old World monkeys, apes, and men) have morphological characters in common that set them apart from the prosimians. One of the most distinct features is their relatively larger brain.

The New World monkeys comprise two groups, the small Callitrichidae and the larger Cebidae. In Callitrichidae, the SIs of limbic structures and brainstem components are close to the simian averages (Fig. 11). In all species of this family, however, NEO and STR have much lower SIs than in simians on the average. *Cebuella*, which is the most primitive member of Callitrichidae [Hershkovitz, 1977], has the least-developed NEO, STR, and DIE of simians (*Alouatta* excluded).

In Cebidae, the mean EI is about 1.5 times higher than in Callitrichidae. Cebidae also show, in contrast to Callitrichidae, higher SIs of the strongly progressive brain components. Among Cebidae there is, however, a great interspecific variability (Fig. 12), indicating occupancy of a wide variety of niches. Whereas in Callitrichidae the coefficients of variation (CV) vary between 6.2% (MES) and 16.6% (BOL), they range from 14.6% (SEP) to 46.9% (BOL) in Cebidae (Table 4). In fact, Cebidae exhibit a great variability in size, morphology, and behavioral adaptations. *Alouatta*, with very low SIs, and the lowest SI of NEO of the simians, differs from the other Cebidae by its special gut adaptations (large stomach and intestines) [Kuhn, 1964].

The differences in EIs and SIs may be related to differences in home range size. Cebidae generally occupy larger home ranges than Callitrichidae [Eisenberg, 1979; Clutton-Brock and Harvey, 1980]. Therefore it may be assumed that Cebidae need to store more environmental information than Callitrichidae. A similar trend can be observed within the Cebidae. *Lagothrix* and *Cebus albifrons*, with the best-developed brains in this family (Fig. 12), occupy a much larger home range than *Alouatta* despite their larger body size [see Eisenberg, 1979].

Old World monkeys, which are among the most successful of all primates, have a higher mean EI than New World monkeys; this is mainly because of the high mean SI of the NEO. In contrast, all other parts have mean SIs that do not surpass the

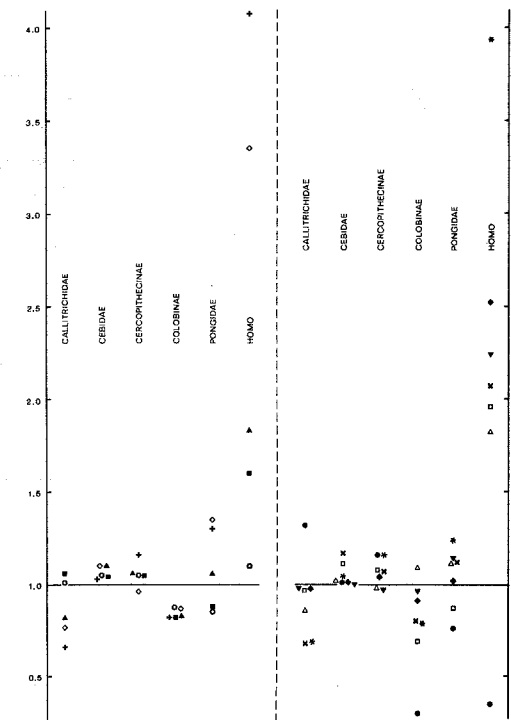


Fig. 11. Size indexes (mostly means) of the brain components in simian taxonomic units. The mean SI of each component in the 26 nonhuman simian species is uniformly set at 1. On the left, the five fundamental brain components; on the right, the seven telencephalic components. Symbols as in Figures 7-9.

means of either Callitrichidae or Cebidae (Table 4). The relatively high CVs indicate a great variability in quantitative brain composition, reflecting extensive radiation. In fact, in addition to their widespread geographical distribution, they are very varied in morphological, ecological, and locomotor adaptations.

Cercopithecoidea are divided into two subfamilies, the Cercopithecoinae and the Colobinae. The two groups differ in a number of features, such as digestive specializations. Cercopithecoinae have cheek pouches, and Colobinae have sacculated stomachs. Cercopithecoinae are more numerous and widely distributed than Colobinae. Most of the brain components of Cercopithecoinae have SIs that surpass the means of simians. The most significant increase in SIs is reached by the NEO (Fig. 11).

Miopithecus, the smallest species of Cercopithecoidea, has one of the highest EIs of the nonhuman

primates, and the SIs of all its brain components (with the exception of the BOL) surpass the mean SIs of the total sample of nonhuman simians. One possible explanation for the high indexes is a secondary reduction in BoW (see below). Although *Miopithecus* shows, particularly in cranial morphology, the basic characteristics of *Cercopithecus*, it has a somewhat neotenuous appearance and resembles the juvenile stages of the larger species [Verheyen, 1962]. Marked differences exist even among various species of the genus *Cercopithecus*. Differences in quantitative brain composition certainly correspond to the exceptional niche diversification of this widespread and abundant genus of African monkeys. This group, which occupies various grades of biotopes, shows a fair amount of dietary, locomotor, and behavioral variation [see Napier and Napier, 1967]. Chromosome number,

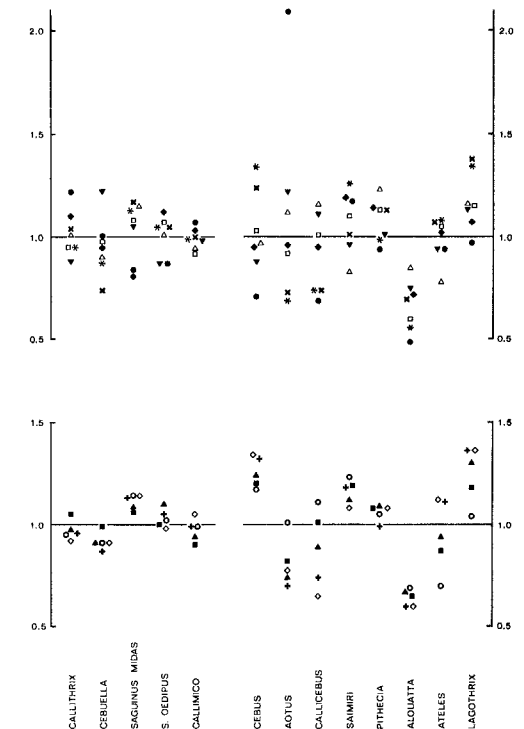


Fig. 12. Size indexes of the brain components in the New World simian species. The mean SI of each component in each of the two families is uniformly set at 1; ie, on the left for Callitrichidae, on the right for Cebidae. In the lower part, the five fundamental brain components; in the upper part, the seven telencephalic components. Symbols as in Figures 7-9.

too, is strongly variable within this group [Delson and Andrews, 1975]. *Erythrocebus* inhabits the open savannas and has a higher SI of the NEO than the average of Cercopithecinae, but the SIs of its motor centers, such as CER and STR, are below the mean.

Cercopithecus, *Miopithecus*, and *Erythrocebus* are considered by some authors [see Szalay and Delson, 1979] a tribe, the Cercopithecini [Burnett, 1828]. The second tribe, the Papionini (*Macaca*, *Cercocebus*, and *Papio*), despite its wide distribution, seems to be much more uniform in brain size and quantitative brain composition. In the brain of the almost wholly arboreal *Cercocebus albigena*, the SIs of most components are higher than in the other species. The most terrestrial species, *Papio anubis*, has the lowest SIs of motor centers of the three Papionini in our sample but the highest SI of BOL of the Old World simians.

Colobinae, the second subfamily of Old World monkeys, have lower average EIs and, with the exception of HIP, lower SIs than the Cercopithecinae and than our total sample of simians (Fig. 11). Most of the morphological features postulated for an early catarrhine are found in Colobinae [Delson, 1975; Delson and Andrews, 1975]. The external morphology of its brain also shows several conservative features [Radinsky, 1974; Falk, 1978]. Hence, despite the great expansion of the digestive tract, characteristic of folivores, which may contribute to the increase in BoW and resultant decrease in indices, its quantitative brain composition seems to indicate a primitive condition.

Apes reach the largest mean EI among nonhuman primates mainly because of the very high SIs of CER and NEO (Fig. 11). Although it shares the basic morphologic character of modern great apes [Szalay and Delson, 1979], *Hylobates* is strongly specialized for an arboreal existence. The gibbons are the most skilled brachiators of all primates, and the SIs of the motor centers (STR, CER) in *Hylobates* are very high and, for CER, the highest among nonhuman primates (Table 3). The relatively high SIs of all brain components (except for olfactory structures) may be partly attributed to a reduction in BoW. Just as the large digestive organs in specialized folivores [Hladik, 1967] contribute to an increase in BoW with no comparable increase in BrW, a reduction in heavier structural material, eg, compact bones and muscles, as found in hang feeders, may contribute to a decrease in BoW with no comparable decrease in BrW [see Ripley, 1979].

Andrews and Groves [1976] have suggested that ancestral Hylobatinae were larger than the modern forms.

Of the nonhuman primates, *Pan* has the highest EI and, with the exception of *Miopithecus*, the highest SI of the NEO. For *Gorilla*, the large BoW, as found for other folivorous species too, seems to be the main reason for the enormous differences between the indices of *Pan* and *Gorilla* (see below).

Summary. The relative size of the total brain (EI values) and its parts (SI values) is lowest in Insectivora and progresses via Scandentia, prosimians, and nonhuman simians, finally to man, who in nearly all parts (except olfactory structures) reaches by far the highest values. Tree shrews deviate with their markedly higher mean SI for the mesencephalon from the general trend of prosimians and nonhuman simians (Fig. 7). Among prosimians, Lemuriformes are more diversified in brain size and quantitative brain composition than Lorisiformes. In *Tarsius* (Tarsiiformes), the EIs and SIs are close to the prosimian average. Among New World simians, the mean SIs of the strongly progressive brain parts are higher in Cebidae than in Callitrichidae. Of the Old World monkeys, Cercopithecinae in general have higher EIs and SIs than Colobinae. Pongidae are characterized by an exceptionally high SI of the neocortex compared with the other brain parts.

Dwarfism and Gigantism

The EIs and nearly all the SIs deviate for some particular species from closely related species in a similar way. We believe that the reasons for such uniform deviation is not due to the brain itself but rather to the BoW. In *Daubentonia* and *Miopithecus* (and *Hylobates*?, see above), the body appears to be "too small" in relation to the brain, whereas in *Gorilla* the body appears to be "too large."

Deviating species have also been found in bats. *Pteropus lylei* deviates from the Pteropodidae in having especially high SIs, and *Hipposideros commersoni* from the Hipposideridae, by especially low SIs [Stephan et al, 1981b]. From the larger number of bat species, it could be shown that such deviations are lone exceptions within families whose species otherwise have similar SIs. Such deviating species seem to be more frequent in primates. Moreover, since most primate families have only a few species, the slopes are much more affected by such deviating forms than elsewhere.

In *Daubentonia* and *Miopithecus*, the generally high indexes may be considered a consequence of a secondary reduction in BoW. These deviating species may have derived from larger-bodied ancestors. According to Lamberton [1934], the fossil *Daubentonia* (*Chiromys*) *robusta* had longer and more robust bones than the living form. In the case of such "dwarfism" (well known in dogs [Stephan, 1954]), the decrease of the brain and its parts appears to have followed intraspecific slopes (generally close to 0.25 for the total brain; see above), which are distinctly lower than the interspecific slopes. In *Gorilla* and *Hipposideros commersoni*, an enlargement of body size ("gigantism") may have occurred and been accompanied by an increase of the brain and its parts in a similar intraspecific relationship. In this way, animals may have been produced having "too large" a BoW for their BrW and therefore comparatively low EIs and SIs. Such forms of gigantism affect mostly males, and both in *Gorilla* and *Hipposideros commersoni*, strong sexual dimorphism has developed. When males and females are investigated separately, the males have distinctly lower EIs than the females. However, in both species, the EIs of the females are also distinctly lower than the EIs of the closest-related species.

Ecoethological Significance

The assumption that the morphological traits of the brain reflect the ecological niche of a species and that even slight structural differences therefore connote differences in ecoethological adaptations is gaining acceptance by ecologists as well as morphologists. More and more studies have shown that the considerable variations in the relative size of the entire brain and of its major regions (especially of the sensory and motor centers) among different mammalian orders such as Insectivora, Chiroptera, and Primates are paralleled by morphological, physiological, and behavior characteristics that, in turn, are related to environmental features. In fact, many primate groups differing in EIs and SIs not only show vast differences in evolutionary trends and geographical distribution but also have adopted different biological strategies.

Ecomorphological studies on primate brains and brain parts have focused mainly on the trophic niche with its various features such as food type, foraging style, and food distribution [Bauchot and Stephan, 1966, 1969; Clutton-Brock and Harvey,

1980; Baron et al, 1983; Frahm et al, 1984; Matano et al, 1985a,b]; time budgeting of activity [Stephan and Bauchot, 1965; Bauchot and Stephan, 1966, 1969; Stephan, 1969; Baron, 1981]; and type of habitat and home range size [Bauchot and Stephan, 1969; Clutton-Brock and Harvey, 1980]. Attempts to correlate brain size with social behavior were first made by Bauchot and Stephan [1966, 1969].

Activity Periods. One of the most conspicuous aspects of the ecological niche is the timing of an animal's activity period. In the prosimian families Lemuridae and Indridae, which comprise both diurnal and nocturnal species, the EIs tend to be higher in the diurnal forms. The same tendency also appears when the EIs of diurnal and nocturnal species are compared across the whole prosimian suborder, where the diurnal species have, on the average, 1.15 times higher EIs than the nocturnal species (Table 5). However, the EIs of the nocturnal species completely overlap those of the diurnal species. In simians, the only nocturnal species (*Aotus*) has one of the lowest EIs. The diurnal species have, on the average, 1.34 times higher EIs than *Aotus* (Table 6).

The parts of the brain also vary with diurnal or nocturnal habits. The SIs of the visual structures (lateral geniculate body, area striata) are on the average between 1.2 and 1.8 times larger in diurnal species than in nocturnal species. The greatest difference (1.8) was found when the lateral geniculate body of *Aotus* was compared with the average of the diurnal simians [Stephan et al, 1984; Frahm et al, 1984]. Larger mean SIs in diurnal species, in both prosimians and simians, are generally found for the strongly progressive structures (CER, DIE, STR, NEO). Low or only slight differences in the mean SIs are found in the brainstem structures (OBL, MES) and in the limbic structures (SEP, HIP, SCH). In contrast to the visual structures, the mean SIs of the olfactory structures are larger in nocturnal than in diurnal species. In prosimians, the mean SI of the BOL is 1.6 times, and in simians, even 2.1 times larger.

Aotus differs, as already mentioned, from all simians by its scotopic vision. The only part that is better developed in *Aotus* than in any other simian is the BOL. Most other indexes are far below the average for Cebidae except the size indexes of the OBL, the HIP, and the SCH (Fig. 12).

TABLE 5. Average Indexes and Coefficients of Variation of Total Brains (TBR) and Brain Structures in Prosimian Ecoethological Groups (based on the Tenrecinae reference line)

	n(s)	TBR	OBL	MES	CER	DIE	TEL	BOL	PAL	SEP	HIP	SCH	STR	NEO
Activity timing														
Nocturnal	14	4.01	1.56	2.70	4.55	5.20	4.69	0.58	0.88	1.87	2.81	3.40	5.58	19.9
		30.4	17.7	22.0	29.7	32.6	33.1	48.7	35.7	24.2	27.2	36.5	44.3	33.5
Diurnal	4	4.62	1.60	2.94	5.30	6.24	5.47	0.36	0.75	1.89	3.05	3.79	6.80	23.2
		22.1	22.7	20.0	23.4	17.6	23.2	54.6	29.9	19.6	17.8	20.2	21.9	26.0
Dietary types														
Insectivores (grade 1)	6	4.84	1.68	3.08	5.24	6.29	5.78	0.65	1.05	2.07	3.08	3.83	6.98	25.2
		28.3	20.6	18.1	33.4	32.2	29.6	52.0	33.4	25.7	34.6	46.6	47.1	25.0
Frugivores (grade 2)	5	3.69	1.59	2.63	4.33	4.56	4.24	0.68	0.90	1.92	2.79	3.21	4.88	17.4
		14.5	7.6	15.3	15.6	16.9	16.0	17.9	15.9	11.4	14.5	21.8	17.0	19.4
Folivores (grade 3)	7	3.88	1.46	2.55	4.54	5.31	4.52	0.31	0.65	1.67	2.74	3.40	5.59	19.1
		30.8	21.5	25.9	28.8	27.8	33.5	48.2	31.9	23.6	20.6	22.0	33.5	36.1
Generalist folivores (grade 3a)	2	5.44	1.89	3.41	6.21	7.17	6.47	0.52	0.94	2.20	3.46	4.35	7.97	27.7
		12.6	0.7	9.5	18.1	4.8	14.5	10.9	0	8.1	13.5	6.7	13.2	20.0
Nocturnal strict folivores (grade 3b)	3	2.89	1.28	2.04	3.54	4.07	3.25	0.26	0.51	1.38	2.33	2.88	3.98	13.5
		14.2	9.2	16.5	12.1	20.9	14.5	26.6	12.9	13.4	11.6	9.7	15.2	17.3
Diurnal strict folivores (grade 3c)	2	3.81	1.31	2.46	4.39	5.32	4.48	0.19	0.56	1.59	2.65	3.24	5.64	18.8
		0.6	18.4	6.3	4.4	4.9	2.5	14.9	6.4	5.8	2.9	21.0	5.4	2.2
Locomotor types														
Arboreal quadrupeds	6	4.71	1.73	2.94	5.54	6.03	5.53	0.71	1.04	2.15	3.28	4.12	6.93	22.9
		36.5	15.4	17.8	33.9	42.4	39.5	29.5	22.5	21.5	29.4	43.3	51.7	40.7
Vertical clingers and leapers	9	3.73	1.51	2.71	4.38	4.94	4.29	0.38	0.67	1.64	2.67	3.15	4.93	18.6
		21.5	19.6	26.0	19.8	17.3	23.3	71.8	41.0	19.2	15.7	15.1	21.1	26.8
Slow-climbing quadrupeds	3	4.26	1.44	2.50	4.06	5.71	5.23	0.60	1.00	2.03	2.64	3.25	6.48	22.5
		5.8	16.2	5.5	1.2	13.6	6.3	30.6	11.4	16.1	27.3	21.3	15.5	10.9

Abbreviations of structures as in Table 3; n(s), number of species.

TABLE 6. Average Indexes and Coefficients of Variation of Total Brains (TBR) and Brain Structures in Simian Ecoethological Groups (based on the Tenrecinae reference line)

	n(s)	TBR	OBL	MES	CER	DIE	TEL	BOL	PAL	SEP	HIP	SCH	STR	NEO
Activity timing														
Nocturnal (<i>Aotus</i>)	1	6.04	1.96	2.93	5.60	6.50	7.72	0.17	0.78	1.97	3.02	3.59	8.45	36.6
Diurnal	25	8.09	1.82	3.46	6.65	7.94	10.8	0.08	0.76	2.03	2.64	2.93	9.99	51.2
		26.2	17.6	18.3	26.3	21.9	28.4	43.2	22.1	14.6	26.0	23.3	30.1	28.4
Dietary types														
Frugivores (grade 2)	21	8.33	1.92	3.64	6.78	8.24	11.1	0.09	0.81	2.09	2.68	3.04	10.4	53.2
		26.0	14.2	13.7	26.3	20.7	28.5	34.6	16.7	12.5	25.2	23.0	29.9	27.7
Folivores (grade 3)	5	6.64	1.46	2.60	5.91	6.36	8.78	0.04	0.54	1.74	2.53	2.58	7.95	39.8
		19.2	14.1	13.4	23.4	10.8	20.3	57.6	8.1	13.6	29.3	19.2	9.7	19.5
Locomotor types														
NW arboreal quadrupeds	10	6.96	2.00	3.71	6.12	7.66	8.93	0.10	0.81	2.05	2.56	2.98	9.02	43.9
		29.5	10.9	11.0	28.5	22.3	32.0	33.2	11.4	11.7	17.7	13.2	34.2	32.2
NW semibrachiators	3	8.49	1.56	3.22	7.44	8.39	11.3	0.07	0.79	1.92	2.52	2.77	12.2	52.5
		36.9	24.4	29.0	37.9	32.6	38.1	34.6	31.1	20.8	21.5	20.2	32.6	39.9
OW arboreal quadrupeds	4	9.31	2.02	3.93	6.73	8.98	12.6	0.11	0.94	2.23	2.73	3.13	11.3	60.8
		17.1	13.8	17.0	16.7	18.9	17.7	27.5	15.1	18.9	16.0	27.9	25.9	20.1
OW terrestrial quadrupeds	3	8.54	1.80	3.16	5.83	7.47	11.9	0.08	0.67	1.94	2.45	2.54	9.57	55.9
		7.9	15.3	8.5	2.3	10.6	9.3	57.3	10.4	10.4	12.7	4.3	6.7	11.3
OW semibrachiators	3	6.59	1.60	2.79	5.72	6.58	8.72	0.03	0.53	1.84	2.88	2.83	7.93	40.2
		11.0	4.4	11.6	7.2	13.0	11.7	78.1	4.8	13.2	25.7	18.0	13.6	12.7
OW true brachiator (<i>Hylobates</i>)	1	10.7	2.02	3.69	9.92	10.6	14.4	0.05	0.83	2.36	5.00	5.26	14.8	66.5
OW modified brachiators	2	9.89	1.33	2.67	8.47	7.24	13.5	0.07	0.58	1.89	1.93	2.44	9.32	60.9
		21.7	17.0	20.7	6.4	18.7	24.8	20.2	7.3	10.9	16.5	13.6	22.5	27.6
Specialized bipedal (<i>Homo</i>)	1	30.1	2.01	5.49	22.1	14.4	43.4	0.03	1.49	5.10	4.81	6.61	20.6	199.2

Abbreviations of structures as in Table 3; n(s), number of species.

Feeding Ecology. Differences in feeding ecology are, to a certain extent, reflected by differences in the EIs and SIs. A classification according to the trophic behavior of the species described here was used in studies on the relative size of the main olfactory bulb [Baron et al, 1983] and of the area striata [Frahm et al, 1984]. This classification, which was proposed by Hladik [1975, 1979], contains three dietary grades characterized by particular food preferences: insectivorous and carnivorous species complementing their diet with fruits and gum (grade 1); frugivorous forms, including gum and seed eaters that consume some animal protein (grade 2); and folivores, which eat mainly leaves as well as shoots and flowers with a supplement of various amounts of fruit (grade 3). A detailed description of the species and their dietary habits is given in the publications cited above.

Prosimians. The mean EI of folivorous (grade 3) prosimians is similar to that of frugivorous species but markedly lower than that of insectivorous and carnivorous forms (Table 5). Within folivores, the relatively high CV indicates that there must be marked differences in their feeding strategies. Folivorous forms cover almost the whole range of EIs found among prosimians. The nocturnal *Lepilemur* has the lowest, and the diurnal *Lemur fulvus* the highest EI (next to the exceptional case of the insect-eating *Daubentonia*). Of the diurnal folivores, the EIs of Lemuridae differ from those of Indriidae. Thus, within the folivorous prosimians, three groups can be distinguished: the diurnal Lemuridae (*Lemur*, *Varecia*; grade 3a), with the largest mean EI; the diurnal Indriidae (*Propithecus*, *Indri*; grade 3c), with an intermediate value; and the nocturnal *Avahi* and *Lepilemur* (grade 3b), with the lowest EI (see Table 5). All the major brain parts also show high variability, and the SIs have trends similar to those of the EIs.

There are major differences in the behavioral ecology of the various folivorous species. According to Hladik [1979], *Lemur* can adapt more easily to changes in diet than, for instance, *Indri* and *Propithecus*, which are more specialized in their diets. The greater behavioral plasticity finds its expression in the higher indices, especially of the strongly progressive structures, as, eg, the NEO, whose mean SI is about 1.5 times larger in diurnal folivorous Lemuridae than in the diurnal Indriidae. Similarly, other brain structures, such as area striata [Stephan, 1969; Frahm et al, 1984], ventral pons [Matano et al, 1985a], thalamus and subthalamus

[Baron, 1979], and corpus geniculatum laterale [Stephan, 1969; Baron, 1981], have been reported to have higher SIs in diurnal Lemuridae (grade 3a) than in diurnal Indriidae (grade 3c).

The ecological niche of the nocturnal leaf eater *Lepilemur* seems to be less complex than that of other folivorous species: *Lepilemur* occupies a very small home range and shows little dietary diversity [Clutton-Brock and Harvey, 1977, 1979b], and it therefore needs to process less varied information. This species feeds mainly on the tough foliage of trees commonly found in the forest [Charles-Dominique and Hladik, 1971; Hladik, 1979]. *Lepilemur* belong to what Ripley [1979] calls the "strict folivores." In the *Avahi* subspecies, which are the other nocturnal leaf eaters, the SIs of the progressive brain structures are slightly higher than in *Lepilemur*. We could not find any report on the exact feeding strategy of *Avahi*. Unlike *Lepilemur*, which generally live in spring bush forests [Hladik and Charles-Dominique, 1974], *Avahi* is found in a more complex habitat, such as rain forests.

All frugivorous (grade 2) prosimians are nocturnal forms. The EIs, and the SIs of the progressive brain parts, are less variable than in folivorous forms, suggesting that niche complexity is more similar among frugivores. Most brain parts have, on the average, similar SIs in frugivores and in folivores. The only parts with higher SIs in frugivores are the BOL (2.2 times) and the PAL (1.4 times). However, when only nocturnal forms are compared, the SIs of all major brain parts, and particularly those of the strongly progressive structures, are on the average higher in frugivores than in folivores. Frugivores tend to occupy larger home ranges than folivores [Clutton-Brock and Harvey, 1977]. Since fruit trees are usually widely dispersed, frugivorous species have to travel over ranges exceeding those of nocturnal leaf eaters of similar size. Locomotion and orientation in a complex environment require well-developed spatial memory and learning capacities in addition to efficient sensory capacities, such as nocturnal vision and olfaction.

Each of the insectivorous and carnivorous (grade 1) prosimians practically represents a different type by both its ecoethological adaptations and SIs. The generally wide range of variation of EIs and SIs found for the grade 1 species is mainly due to *Daubentonia*. Although all indexes are high, possibly because of a miniaturization of the BoW, as was already pointed out, the EIs of the various brain parts of *Daubentonia* deviate to different ex-

tents from the prosimian means (Fig. 10). The highest differences are found in centers related to the motor system (CER, STR, DIE) and in the BOL. *Daubentonia* locates the larvae of wood-boring insects by smelling and hearing the noise produced by the insects beneath the bark [Hladik, 1979].

Tarsiers form the second type of grade 1 prosimians. They feed exclusively on animal prey detected mainly by sight and hearing [Niemitz, 1979, 1984]. Olfaction plays a minor role. Next to Indriidae, tarsiers have the lowest SIs for the BOL and PAL of the prosimians [Stephan, 1966; Baron et al, 1983]. In contrast, their visual centers are well developed [Stephan, 1969; Baron, 1981; Stephan et al, 1984; Frahm et al, 1984]. In *Tarsius*, the MES, including the anterior colliculi, which are important visual structures, has the highest SI of all brain parts, when compared to other prosimians (Fig. 10).

In Cheirogaleidae, the EI, and most SIs of the predatory *Microcebus*, are similar to those of the two frugivorous forms. However, in *Microcebus*, the SI of the BOL is slightly lower and that of the NEO is slightly higher. According to Hladik [1979], hearing and sound detection of movements seem to be more important than olfaction in the predation strategy of *Microcebus*. *Galagoides* resembles *Microcebus* in its diet, feeding strategy, and ecology. There are, however, differences in their microenvironments. While *Microcebus* lives in the dense undergrowth, *Galagoides* lives at the top of the canopy. Furthermore, the proportion of dietary animal protein is higher in *Galagoides* than in *Microcebus* [Hladik, 1979]. Next to *Daubentonia*, *Galagoides* has the highest EI among insectivorous prosimians, and all SIs are higher in *Galagoides* than in *Microcebus*. This is also true of the olfactory structures (BOL, PAL), which reach the highest values among prosimians in *Galagoides* (next to *Daubentonia*). Most SIs are higher in *Galagoides* than in the frugivorous Galaginae.

The last group of grade 1 prosimians is formed by the insectivorous Lorisinae. Again, as was seen in the Galaginae and the Cheirogaleidae, the insect-eating forms (*Loris*, *Nycticebus*) have higher EIs than the frugivorous *Perodicticus*, and the SIs of most of the strongly progressive structures are higher too.

Loris and *Nycticebus* differ from *Galagoides* in that they catch prey generally avoided by other insect eaters because of its repugnant smell and taste [Elliot and Elliot, 1967; Charles-Dominique

and Bearder, 1979; Hladik, 1979]. The high intensity of the normally repulsive odors of the prey may explain why the BOL in the two species is only two-thirds the size of that of other prosimians that hunt insects by smell. Although olfaction predominates in the detection of prey, vision is an important source of information about the environment. In fact, the SIs of the striate area [Frahm et al, 1984] and the corpus geniculatum laterale [Stephan et al, 1984] are highest among night-active prosimians (except for *Tarsius*).

Simians. Among nonhuman simians, the mean EI and all mean SIs of the frugivorous species are higher than those of the folivorous forms (Table 6). The greatest difference is in the BOL: its mean SI is more than twice as large. There are clear parallels with prosimians in this respect. In all simian families there is a clear relationship between the indexes and the dietary types. *Alouatta*, which averages the greatest leaf consumption of all neotropical primates [Eisenberg, 1979], has the lowest SIs of all Cebidae for most of the brain parts (Fig. 12). Among the paleotropical Cercopithecidae, the predominantly folivorous (grade 3) Colobinae have, as already mentioned, much lower mean EIs and SIs than the mainly frugivorous (grade 2) Cercopithecinae (Table 4). Similar results have been reported by Bauchot and Stephan [1969] and by Clutton-Brock and Harvey [1980]. One of the most extreme folivores, *Nasalis* [Kay, 1978], has the smallest SIs for most of the brain parts (Table 3).

Finally, in Pongidae, the EIs and nearly all SIs of the predominantly frugivorous forms (*Pan*, *Hyllobates*) are clearly higher than those of the predominantly folivorous *Gorilla* (Tables 1, 3).

Locomotor Types

Prosimians. In prosimians there are three different locomotor types: vertical clingers and leapers [Napier and Walker, 1967] (VCL: *Lepilemur*, Indriidae, Galaginae, *Tarsius*); arboreal quadrupeds (AQP: Cheirogaleidae, Lemurinae, *Daubentonia*); and slow-climbing quadrupeds (SCQ: Lorisinae) (see Table 1). General motor activity, particularly locomotion, is a major aspect of an animal's interaction with its environment and should parallel differences in brain structure.

VCLs have on the average the lowest EIs and SIs. Their mean EI is surpassed by that of SCQs (1.15 times) and by that of AQPs (1.3 times). The most conspicuous differences in the mean SIs are found in the olfactory structures (BOL 1.9 and 1.6

times, respectively). Large differences also exist in the STR, whose mean SIs are 1.4 times larger in AQP and 1.3 times in SCQs than in VCLs. However, the mean values may be misleading because of high intragroup variability (Table 5). Indeed, the species within these locomotor groups are very heterogeneous with respect to their activity timing and diet (Table 1).

That brain parts have higher mean SIs in AQP than in VCLs (Table 5) is mainly due to the high SIs in *Daubentonia* and *Lemur*. However, among the frugivorous types, the VCLs (*Galago*, *Otolemur*) have about 1.2 or 1.3 times larger SIs of most of brain parts than the AQPs (*Cheirogaleus*). Despite a number of similar morphological and behavioral traits [Charles-Dominique and Martin, 1970], Galaginae and Cheirogaleinae differ in their locomotor abilities. *Cheirogaleus* species are less agile leapers than Galaginae [Cartmill, 1974]. The locomotor system of the Galaginae, in which leaping plays an important role, permits them to exploit a wider range of habitats, in particular those with discontinuous tree cover [Charles-Dominique and Bearder, 1979]. Among frugivores, the slow-climbing *Perodicticus* (SCQ) has a very low SI of the CER but the highest SI of the STR, which is 1.1 times larger than in VCLs and 1.4 times larger than in AQPs. According to Walker [1969], *Perodicticus* has a frequency of leaping that is, for all practical purposes, zero. The stealthy, slow-climbing locomotion in the upper layers of the canopy are made possible by very finely tuned movements.

In the diurnal folivores, the EI is 1.4 times larger in AQPs than in VCLs, and all the major brain parts have higher SIs. The AQP folivores (Lemurinae) are capable of more diversified movements than the VCL Indriidae, the other group of diurnal folivores [Walker, 1979]. When leaping from trunk to trunk, VCL Indriidae maintain a more or less vertical posture, usually following a flat trajectory [Petter, 1962; Walker, 1979]. The estimation of distance is, in such cases, relatively unimportant [Pariente, 1979]. In AQP Lemurinae, the body posture during leaps is more inclined in the direction of the leap. Lemurinae also leap from a higher to a lower level, a task demanding precise evaluation of the distance [Walker, 1979; Pariente, 1979] and good muscle coordination. The more complex motor and sensory patterns of Lemurinae are paralleled by better-developed motor and sensory centers, such as the NEO, STR, DIE, CER, and MES (Table 5, grade 3a). The corresponding

mean values in diurnal folivorous VCL Indriidae are much lower (Table 5, grade 3c).

Although they are both insectivorous AQPs, *Daubentonia* and *Microcebus* differ in their locomotor adaptations. The most distinctive feature in *Daubentonia*, which has the highest SIs throughout, is its rather slow quadrupedal locomotion in any position and direction [Walker, 1974, 1979]. Its slow movements during its search for prey may also be considered a cryptic strategy to avoid the attention of predators, similar to the slow-climbing habit characteristic of lorises [Charles-Dominique and Bearder, 1979]. The high SI of the motor center suggests that its slow locomotion is accomplished by finely tuned movements. The locomotion of *Microcebus*, which has much smaller SIs, is characterized by rapid displacements in a jerky fashion [Walker, 1979]. Differences are also found among the VCL insectivorous forms *Galagoides* and *Tarsius*. These differences are particularly pronounced in the sensory structures. The SI of the BOL is 5.5 times larger in *Galagoides* than in *Tarsius*; the SI of the visual cortex, on the other hand, is 1.3 times larger in *Tarsius* than in *Galagoides* [Frahm et al., 1984]. Typical motor structures, such as the CER and the STR, are also better developed in *Galagoides* than in *Tarsius* (ie, 1.3 and 1.7 times, respectively). Among the VCLs, *Galagoides* seems to be the most versatile locomotor type. The animals progress by running, walking, climbing quadrupedally, and leaping between supports [Walker, 1979]. *Tarsius* and *Lepilemur* are the most typical VCLs.

Simians. Two locomotor types can be distinguished in New World simians: the arboreal quadrupeds (NAQ: Callitrichidae, Cebidae except NSBs) and the semibrachiators (NSB: *Alouatta*, *Ateles*, *Lagothrix*). Despite considerable intragroup variation, both groups show particular trends in their mean EIs and the SIs of various brain parts. The mean EI is lower in NAQs than in NSBs. The two brainstem components OBL and MES are the only parts that have slightly higher mean SIs in NAQs than in NSBs. Size differences between limbic structures are negligible. The DIE, the NEO, and the motor centers (CER, STR) have higher mean SIs in NSBs than in NAQs. Among NAQs, the small Callitrichidae differ from the larger Cebidae in limb proportions and predominant motor patterns. According to Erikson [1963], Callitrichidae, whose hind limbs are approximately 25%

longer than their forelimbs, are essentially quadrupedal branch runners and springers. They still retain jerky movements, which seem to be characteristic of a primitive locomotor system. Apart from the longer and more powerful hind limbs and longer digits, their postcranial skeleton probably differs little from that of the postulated ancestral type [Hershkovitz, 1977]. The quadrupedal Cebidae form an intermediate group between the quadrupedal Callitrichidae and the semibrachiating Cebidae in that they tend toward more specialized arboreal locomotor types. In the quadrupedal Cebidae, the forelimbs become increasingly important through the acquisition of longer arms and grasping hands with very mobile fingers. The increasing motor ability, indicating greater complexity in the ecological niche, is reflected by the higher EIs and SIs, particularly of the strongly progressive parts.

The tendencies observed in the quadrupedal Cebidae become more accentuated among the semibrachiating New World monkeys, which have even longer arms, are better climbers, and are adapted, to some extent, to brachiation. *Ateles* and *Lagothrix* have high EIs and SIs, particularly of the motor structures. In contrast, the indexes of *Alouatta* are surprisingly low, and in many cases (eg, NEO) the lowest of all neotropical simians. It was already noted that *Alouatta* is mainly a leaf eater. In addition, its classification as a semibrachiator does not seem to meet unanimous agreement. According to several authors, *Alouatta* is a quadrupedal climber rather than a brachiator [Ankel, 1962, 1963; Grand, 1968; Knussmann, 1967; Schön, 1968]. Its leaping ability also seems to be very poor. Finally, its tail, compared with that of the other NSBs, is primitive and serves primarily as a supporting organ for climbing and body suspension [Ankel, 1962]. The tail of Atelinae, in contrast, can act as an independent additional organ for swinging and propulsion and thus increases their locomotor repertoire [Hershkovitz, 1977].

A different trend is observed among Old World simians. Compared with arboreal quadrupeds (OAQ: *Cercocebus*, *Cercopithecus*, *Miopithecus*) and terrestrial quadrupeds (OTQ: *Macaca*, *Papio*, *Erythrocebus*), the semibrachiators (OSB: Colobinae) always, with the exception of some limbic structures, have the lowest SIs. On the other hand, the only true brachiator (OTB: *Hylobates*) of our sample has consistently, apart from the olfactory structures, the highest SIs as compared with the

averages of all locomotor types (*Homo* excluded; Table 6). Among Old World quadrupeds, the terrestrial forms (OTQ) have lower mean SIs than the arboreal species (OAQ).

Differences in motor patterns and brain indexes are paralleled by differences in limb proportions. The Cercopithecinae, which are mainly arboreal quadrupeds, have shorter arms than legs. They are powerful leapers. In mainly terrestrial species, the arms tend to become as long as the legs [Hershkovitz, 1977; Gabis, 1960]. The semibrachiation attributed to Colobinae by Napier [1963] is questioned by Hershkovitz [1977]. In fact, Colobinae exhibit only few characters of brachiators [Knussmann, 1967] and are probably mainly climbers. Their forelimbs and hind limbs are nearly equal in length. According to morphological features, OSBs are certainly less specialized than NSBs.

Hylobates is one of the most typical brachiators. The very high SIs, especially in those brain structures that are or that contain important motor centers (CER, STR), imply a greater complexity of motor patterns than found in other nonhuman primates. Similarly, high indexes have been found for the diencephalic motor centers [Baron, 1979] and for the ventral pons [Matano et al., 1985a]. Brachiation, which constitutes the major means of arboreal locomotion [Napier and Napier, 1967], may account in part for the high SIs of the motor centers. The generally very high SIs of the brain centers, however, may be caused by other factors too, as eg, by decrease in body size (see above).

Despite their moderate brachiation, the great apes exhibit even more extreme adaptations than *Hylobates* in the trunk and limbs [Erikson, 1963], but, by becoming large-bodied, they have developed into slower arboreal and terrestrial forms (OMB: *Pan*, *Gorilla*). Chimpanzees are as much terrestrial as arboreal, but, although they apparently prefer to knuckle-walk [Tuttle, 1975], they frequently employ arm swinging [Goodall, 1965; Reynolds and Reynolds, 1965]. *Gorilla*, the other modified brachiator, is mainly terrestrial [Schaller and Emlen, 1962]. The change in locomotor pattern from true brachiators to modified brachiators is paralleled by changes in quantitative brain composition. Although the SI of the NEO is about 4.5 times larger than the SI of the STR in *Hylobates*, it is more than six times larger in the two great apes.

Summary

Activity periods. Nocturnal primates have, on the average, lower SIs in the visual components but

higher SIs in the olfactory parts, whereas the limbic and brainstem structures are less affected. There is, however, a large overlap in the indexes for individual species.

Feeding ecology. Differences in the indexes are related to a certain extent to differences in the feeding ecology. There is a wide range of variation of EIs and SIs in insectivorous and carnivorous (grade 1) prosimians, suggesting a great diversity of behavioral patterns and strategies. In frugivorous (grade 2) prosimians, the indexes are less variable than in other dietary types. Differences in EIs and SIs among folivorous (grade 3) prosimians are paralleled by marked differences in food preferences and foraging strategies. In all simian families, folivores usually have lower indexes than frugivorous forms.

Locomotor types. Despite an appreciable overlap of the indexes of prosimian locomotor types in many cases, there is a good relationship between the relative size of brain parts and the particular motor pattern. Species capable of more diversified movements (eg, arboreal quadrupedal folivores) (which all include motor components to a considerable degree) than species with less complex locomotor abilities (eg, vertical clinging and leaping folivores). Among the New World simians, the increasing importance of forelimbs is paralleled by higher EIs and by higher SIs of the strongly progressive brain parts. Among Old World simians, arboreal quadrupedal forms (OAQs), with shorter forelimbs, have higher SIs than terrestrial quadrupedal forms (OTQs) and semibrachiators (OSBs). However, the true brachiator (OTB) *Hylobates*, with extremely long forelimbs, and the modified brachiators (OMBs), with even more specialized forelimbs, have the largest strongly progressive brain parts (especially the cerebellum) of nonhuman primates.

CONCLUSIONS

Principles

This comparison of the size of the brains (brain weight, BrW) and brain parts is based on indexes that directly reflect size differences (ratios) in species of equal body weight (BoW). Such indexes, like any other values on relative brain size (RBrS) reported by other authors, are estimations which generally are based on the power function intro-

duced by Snell [1892] $BrW = RBrS \times BoW^a$. However, the choice of samples to which this allometric equation is applied has far-reaching consequences. When higher taxonomic units (eg, one or several mammalian orders) make up a given sample, the best-fit line (in log BrW-log BoW plots) has a relatively steep slope. Such a line is then used as the reference line to evaluate RBrS. This method is mathematically clear and apparently indisputable, yields results even if the number of species (with mostly a large BoW range) is restricted, and has therefore gained increasing acceptance in recent publications (eg, by psychologists). However, the obtained RBrS values are often devoid of any biological meaning and may result in such oddities as similarity of relative size values for the total brain and many brain parts in man and in small prosimians, or higher values in small prosimians than in apes.

The inferential sterility with respect to species-specific biological differences, and the fallacy of conclusions, such as mentioned above, are the reasons we do not use higher-order taxonomic groupings. Instead, we use samples of closely related species (genera to families) despite the greater difficulty in determining a satisfactory common slope (which generally is markedly lower than that found for higher-order taxa combinations) and in selecting a reference line. The use of closely related groups of species and a constant slope is not at all new. This was done very early on by Snell [1892a,b] and Dubois [1897]. When using a common slope, the placement of the reference line becomes irrelevant, since the vertical (BrW) distances (ie, the differences in the y intercepts) between species (and thus the ratios between indexes) are unchanged. Our indexes are numerical values of these distances. We prefer to place the reference line through primitive Insectivora (eg, Tenrecinae), but all indexes are freely convertible to other (parallel) reference baselines (eg, through the means of prosimians, simians, man, etc).

Trends

The relative size of some brain parts shows strong differences between individual primate species. When the mean SIs of the various taxonomic units are ranked in an ascending sequence, a series results going from Insectivora, through prosimians, monkeys, and apes, finally to man at the summit. The most progressive brain part is the neocortex. Strongly progressive brain components

are the striatum, cerebellum, and diencephalon. In the same series, a descending sequence is formed by the olfactory bulb (and olfactory paleocortex). Other brain components such as the limbic structures (schizocortex, septum, hippocampus) and mesencephalon progress more slowly, and the medulla oblongata may be considered conservative. Such quantitative changes certainly reflect major trends in brain evolution during phylogeny of primates, even though they do not show the real sequence of events.

The species have radiated to exploit a variety of ecological niches. Differences in relative size of brains and brain parts should reflect differences in ecological and behavioral adaptations. However, simple interrelationships may remain masked by the wide spectrum of factors characterizing the ecological niche. Another factor complicating interpretation of brain size differences in single species is an apparently aberrant body size, which may appear as dwarfism or gigantism. Nevertheless, studies on the differential relative size of brain parts help uncover differences in the complexity and quality of ecological niches.

Because of the great variability of the indexes and the small number of species investigated so far, the results given represent only a first attempt. More data are needed on sufficiently diversified material for a reliable statistical treatment. Unfortunately, it is extremely difficult to obtain data from additional primate species (eg, prosimians from Madagascar), and many families have very few extant species. Therefore, we have deliberately avoided using statistical methods in analyzing our data. The indexes given in Tables 3-6 will enable other researchers to apply such tasks as they consider valid.

Research Needs

From the preceding survey and conclusions, it becomes evident that, to advance our knowledge of brain evolution, future research must focus its interests on three particular problems. One of the major needs has been, and will continue to be, for a more careful selection of appropriate methods to produce biologically meaningful results. Pompous speculations that miss the point, impressive catch phrases that dilute the real problems, mathematical escapades remote from biological realities, and finally the search for a kind of magic slope like 2/3, 3/4, or 1 (isometry) may only hinder real progress in the understanding of relative brain size

and its evolution. The second, but not less urgent need, is to use samples that include as many primate species as possible. Small samples may lead to spurious results. Finally, for a better comprehension of the relationships between structural size and ecoethological features, investigations on more homogeneous structures, such as nuclei and cortical areas, are imperative. Such investigations would also give deeper insights into structural rearrangements due to evolutionary compensations that may have occurred during phylogeny. This was shown in the case of Chiroptera by Baron and Jolicoeur [1980] and Jolicoeur and Baron [1980]. Avenues of exploration have already opened up using more extensive primate material, and investigating size variations of particular nuclei or brain centers, such as the accessory olfactory bulb [Stephan et al, 1982], the amygdala [Stephan and Andy, 1977], the visual structures [Stephan, 1969; Baron, 1981; Stephan et al, 1984; Frahm et al, 1984], the diencephalon [Bauchot, 1979, 1982; Baron, 1979], and the pontocerebellar structures [Matano et al, 1985a,b].

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Input-Output Organization of the Primate Cerebral Cortex

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INTRODUCTION

The cerebral cortex is not a homogeneous structure. A fundamental issue, then, concerns how it should properly be divided. Probably one of the earliest, and simplest, classifications divided the cerebral cortex into two major types: "primary" and "association" cortex [Flechsig, 1920]. "Primary cortex" chiefly encompasses primary motor (MI), somatic sensory (SI), auditory (AI), and visual (VI) areas. Although principally defined in terms of function, certain anatomical criteria were also used in the original definitions of these regions. One of these criteria was architectonic. Each primary cortex corresponds to an area with distinct architectonic features. In the schema of Brodmann [1905, 1909], MI is equivalent to area 4, SI to areas 3, 1, and 2, AI to area 41, and VI to area 17 (Fig. 1). Furthermore, there is some similarity in the cytoarchitectonic appearance of at least the three primary sensory cortices, since each has a prominent fourth layer of granular cells ("konio-cortex"). Another anatomical criterion originally used to define the primary cortices was the pattern of subcortical connections. Motor cortex is the main source of the classical corticospinal tract. Each of the three primary sensory cortices receives relatively direct input from the periphery by way of one of the thalamic sensory relay nuclei.

Initially, "association" cortex was defined, in a negative sense, as comprising all areas of the cerebral cortex other than primary cortex. Behavioral observations suggested that these areas subserved particularly complex operations [Chow

and Hutt, 1953]. Since they did not appear to have direct connections with lower centers in the neuraxis, the association areas were assumed, in the case of post-Rolandic cortex, to receive sensory input by way of the primary sensory cortices, or in the case of frontal association cortex, to influence motor function by way of projections to primary motor cortex. Implicit was the assumption of a series of cortico-cortical connections underlying the functions of these areas, but technical limitations made it impossible to trace the cortico-cortical connections with any degree of confidence. Although there were numerous attempts earlier in this century to subdivide association cortex in terms of architectonics [von Bonin and Bailey, 1947; Brodmann, 1905, 1909; von Economo and Koskinas, 1925; Vogt and Vogt, 1919], these efforts met with limited acceptance in the absence of connective correlations.

Subsequent developments tended to call the simple dichotomy of primary and association cortices into question. In the 1940s and 1950s, physiological studies showed that certain areas of "association cortex" have functional properties that are not dissimilar from those of the traditional primary areas (Fig. 2). Thus, a second somatic sensory representation (SII) was identified in the parietal opercular region of the rhesus monkey, adjacent to primary somatic sensory cortex [Woolsey, 1947; Woolsey and Fairman, 1946]. A second representation of the musculature of the opposite side of the body (MII) was discovered, rostral to primary motor cortex, in the dorsal part of the premotor region (Area 6) [Woolsey et al, 1951]. A second